ستيب STEP: تجميعات متكررة

ملف مراجعة مجاني لاختبار ستيب

القطع الأكثر تكرارًا

STEP Reading — The Most Repeated Passages

40	128
قطعة من الأكثر تكرارًا	سؤال مع الشرح

قطع القراءة اللي تتكرر أكثر في الاختبار، كما وردت في تجميعات المختبرين، مرتبة من الأكثر ذكرًا، ومع كل سؤال الإجابة وسببها من القطعة نفسها.

ملف مجاني من تطبيق ستيب

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انضم لقناة التطبيق على تليجرام (اختبار يومي مجاني): t.me/step_tajmeeat

كيف تستخدم الملف؟ وكيف تحل أسئلة القراءة؟

1 ابدأ من أول قطعة

القطع مرتبة حسب عدد مرات ذكرها في تقارير المختبرين، فأول القطع هي الأكثر تكرارًا في الاختبار. اقرأ القطعة مرة سريعة، ثم حل أسئلتها، ثم صحّح من الإجابات اللي تحتها مباشرة.

2 الفكرة الرئيسية Main idea

أسئلة مثل What is the main idea أو best title: اقرأ أول جملة في كل فقرة. الإجابة الصحيحة تغطي القطعة كلها (أو الفقرة المطلوبة)، والخيارات الغلط تكون تفصيلة صغيرة أو فكرة أوسع من القطعة.

3 التفاصيل Detail

أسئلة مثل According to paragraph و What does the passage say: خذ كلمة مميزة من السؤال (اسم، رقم، سنة) وابحث عنها في القطعة، ثم اقرأ الجملة اللي فيها كاملة. الإجابة الصحيحة غالبًا نفس المعنى بكلمات ثانية.

4 معنى الكلمة Vocabulary

أسئلة closest in meaning to: ارجع للكلمة في الفقرة، واقرأ الجملة قبلها وبعدها، ثم ضع كل خيار مكان الكلمة؛ الخيار اللي يحافظ على معنى الجملة هو الصحيح.

5 الضمير يعود على ماذا؟ Reference

أسئلة The word it / they / his refers to: ارجع للخلف وابحث عن أقرب اسم قبل الضمير يناسبه في العدد (مفرد أو جمع) والمعنى، وجرّب تضعه مكان الضمير.

6 الاستنتاج Inference

أسئلة What can we understand و fact or opinion: الإجابة غير مكتوبة حرفيًا لكنها تُفهم من القطعة بوضوح. ارفض أي خيار يضيف معلومة غير موجودة. والرأي opinion هو كلام يحتمل الاختلاف مثل I believe أو suggest، والحقيقة fact رقم أو حدث يمكن التأكد منه.

تبي تتدرب أكثر؟ في تطبيق «ستيب STEP: تجميعات متكررة» مئات القطع والأسئلة من نفس التجميعات، مع شرح عربي لكل سؤال واختبارات تجريبية ومتابعة لأخطائك.

ستيب STEP: تجميعات متكررة

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ستيب STEP: تجميعات متكررة

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أرقام الصفحات هي الأرقام المكتوبة أسفل كل صفحة. «مرات ذكرها» = عدد مرات ذكر القطعة في تقارير المختبرين، والقطع مرتبة من الأكثر تكرارًا إلى الأقل.

ستيب STEP: تجميعات متكررة

ذكرها المختبرون 164 مرة

4 سؤال

01 التضخم Inflation

(1) Inflation is a general and sustained increase in prices over time. It is calculated by measuring the price of a typical basket of common products that people buy regularly. It tracks how the price of that basket of products changes over time. The rate of inflation is usually stated as a percentage. An annual inflation rate of 2% means that a product that was 100 riyals last year will now be priced 102 riyals.

(2) Inflation affects almost all people. It reduces how much a person can buy because more money is now needed to purchase the same items. High rates of inflation mean that if salaries do not increase at the same rate, people will feel that they are poorer. They will have the same amount of money, but they will not be able to buy the same things. They will purchase less and stop purchasing luxury items and things that do not really need.

(3) Inflation has an effect on businesses too. They often see a drop in sales, because of the lower real levels of consumer spending. In addition, their staff may ask for higher salaries because they need more money to pay for the higher cost of living. This would result in higher costs for businesses and could also cause the business to hike prices even more. This would then push inflation higher again.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

1 تفاصيل

What may cause higher costs for a business?

A an increase in prices.
B higher interest rates.
C higher salaries.
D a fall in sales.

2 تفاصيل

What might people do because of high inflation?

A buy common products.
B buy fewer products.
C spend less money.
D work longer hours.

3 معنى كلمة

The word "purchase" in paragraph (2) is closest in meaning to _____

A buy
B sell
C find
D borrow

4 معنى كلمة

The word "hike" in paragraph (3) is closest in meaning to _____

A report
B improve
C decrease
D increase

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01 الإجابات مع الشرح

1 C higher salaries.

السبب الفقرة 3 تقول إن الموظفين يطلبون رواتب أعلى، وهذا يرفع تكاليف الشركة.

> *their staff may ask for higher salaries ... This would result in higher costs for businesses*

ليه الباقي غلط ارتفاع الأسعار وقلة المبيعات مذكورة في الفقرة، لكنها ليست سبب «التكاليف الأعلى»، والفوائد غير مذكورة أصلًا.

2 B buy fewer products.

السبب الفقرة 2: الناس يشترون أقل لأن نفس المال لم يعد يكفي.

> *They will purchase less and stop purchasing luxury items*

ليه الباقي غلط spend less money فخ: القطعة تقول إنهم يشترون كمية أقل، وليس إنهم يصرفون مالًا أقل؛ بالعكس يحتاجون مالًا أكثر.

3 A buy

السبب purchase = يشتري، والجملة نفسها تشرحها: يلزم مال أكثر لشراء نفس الأشياء.

> *more money is now needed to purchase the same items*

4 D increase

السبب hike prices = يرفع الأسعار، والجملة التالية تؤكد: هذا يرفع التضخم أكثر.

> *could also cause the business to hike prices even more. This would then push inflation higher again.*

ستيب STEP: تجميعات متكررة

02

الحريق: مايك وبن
Fire (Mike and Ben)

ذكرها المختبرون 156 مرة

3 سؤال

(1) Over the roar of the fire, Mike heard Ben shout, "Hurry! The fire is almost on us!" Mike's arms were sore and tired, but he swung his axe even faster. He didn't even stop to wipe the tears from his stinging eyes. The greedy fire kept coming. The more the fire destroyed, the more it wanted. Mike worked shoulder to shoulder with the other smokejumpers to build a firebreak. His only thought was to stop the flaming monster that was raging through the forest.

(2) At last, the smoke jumpers finished the firebreak. If the fire were powerful enough, it would jump over the firebreak that they had worked so hard to make. Then they would have to start all over again. Mike stood motionless, his face black with ash, his shirt wet with sweat. He was too exhausted to move because he had given all of himself to fighting the fire. He turned his head and noticed Ben watching him.

(3) Suddenly all that Ben had taught Mike about proving his bravery was clear. A man was not brave if he did something just to prove his courage. He was brave only when he forgot about himself. Today Mike had showed that he cared very much about the others with whom he was working.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

5 تفاصيل

According to Paragraph (2), Mike was too exhausted to move because _____

A He had sweat all over him.
B He was covered with black ash.
C Ben did not give him much help.
D He had given so much to fight the fire.

6 تفاصيل

According to Paragraph (3), Ben taught Mike that being brave involves _____

A working with proper tools
B avoiding fire hazards
C losing one's courage
D caring for others

7 مرجع الضمير

The pronoun his in Paragraph (3) refers to _____

A Ben
B Mike
C a fireman
D a man

STEP ستيب: تجميعات متكررة

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02 الإجابات مع الشرح

5 D He had given so much to fight the fire.

السبب السبب مكتوب في نفس الجملة بعد because: أعطى كل ما عنده في مكافحة الحريق.

> *He was too exhausted to move because he had given all of himself to fighting the fire.*

ليه الباقي غلط العرق والرماد وصف لحاله وليس سبب التعب، و Ben لم يُذكر أنه قصّر في المساعدة.

6 D caring for others

السبب الشجاعة عند Ben = أن تنسى نفسك وتهتم بالآخرين.

> *He was brave only when he forgot about himself. Today Mike had showed that he cared very much about the others*

7 D a man

السبب الضمير his في الجملة يعود على أقرب اسم مناسب قبله: A man.

> *A man was not brave if he did something just to prove his courage.*

ليه الباقي غلط الجملة قاعدة عامة عن أي رجل، وليست عن Mike أو Ben تحديدًا.

ستيب STEP: تجميعات متكررة

03

ملكية البرامج
Software Ownership

ذكرها المختبرون 120 مرة

1 سؤال

Software Ownership

(1) Most people know that software can be very expensive. However, many people do not know that even though you pay a lot for software, you do not necessarily own it. What happens, then, when you pay for a piece of software? Very few developers grant you actual ownership of a program, even after you've purchased it. Instead of buying the software itself, you are actually paying for a license that grants you permission to install and use the software.

Why a License?

(2) Software developers have good reason for licensing software instead of selling it outright: Piracy-software piracy, which costs developers billions of dollars every year, is the act of copying software without the developer's consent and without paying the developer and then selling or giving away the copies. If you install the program on multiple computers at the same time, that is piracy too. By licensing their products and maintaining some ownership of them, developers can take action against pirates.
Modifications - Most license agreements state that you cannot make modifications to a program's source code. If developers allowed this, it would be an easy matter for others to make changes to a program and then try to claim the modified program as their own.

(3) But even though the developer keeps most rights to a program, you have some rights, too. If the program does not perform as you expected, you have the right to return or exchange it.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

8 What can we understand from the passage about software piracy?

استنتاج

A It is a trivial problem for the software industry.
B It makes lot of money for the software industry.
C It is a bigger issue in the USA than in other countries.
D It is the main reason software companies prefer licensing.

03 الإجابات مع الشرح

8 **D** It is the main reason software companies prefer licensing.

السبب أول سبب يذكره الكاتب لتفضيل الترخيص على البيع هو القرصنة، فالترخيص يسمح للمطورين بملاحقة القراصنة.

> *By licensing their products and maintaining some ownership of them, developers can take action against pirates.*

ليه الباقي غلط القرصنة تكلف المطورين مليارات (ليست مشكلة بسيطة ولا تربحهم)، ولا ذكر لأمريكا.

ستيب STEP: تجميعات متكررة

04 الضغط النفسي Stress

ذكرها المختبرون 113 مرة

6 سؤال

(1) We generally use the word "stress" when we feel that everything seems to have become too much. We are overloaded and wonder whether we really can cope with the demands placed upon us. Anything that poses a challenge or a threat to our well-being is a stress at all, many say our lives would be boring and would probably feel pointless. However, when the stresses affect both our mental and physical health they are bad.

(2) The way we respond to a challenge may also be a type of stress. Part of our response to a challenge is physiological and affects our physical state. When faced with a challenge or a threat, our body activates resources to protect us; to either get away as fast as we can. Or fight. This is known as our 'fight-or-flight' response.

(3) If you are upstairs at home and an earthquake starts, the faster you can get yourself, and anyone else who might be at home, out the more likely you are all to survive. If you need to save somebody's life during that earthquake by lifting a heavy weight that has fallen on them. You will need parts of your body to be activated to give you more strength.

(4) To do this, our body produces larger quantities of the chemicals cortisol, adrenaline and noradrenaline, which trigger it to have a higher heart rate. Heightened muscle preparedness, sweating, and alertness. All these factors help us protect ourselves in a dangerous or challenging situation.

(5) Non-essential body functions slow down, such as our digestive and immune systems when we are in a 'fight-or-flight' situation. All resources can then be concentrated or rapid breathing, blood flow, alertness and muscle use.

(6) Most of us have varying interpretations of what stress is about and what matters, sort of us focus on what happens to us, such as breaking a bone or getting a promotion. While others think more about the event itself. In my opinion. What really matters are our thoughts about the situations in which we find ourselves.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

9 تفاصيل

What does the expression 'fight or flight response' used in paragraph (2) mean?

A Staying in a dangerous situation or facing it

B Getting away from a dangerous situation very quickly

C The ways a dangerous situation changes with stress

D The ways a person's body reacts to a dangerous situation

ستيب STEP: تجميعات متكررة

10 تفاصيل

According to Paragraph (3), what is the best advice in the event of an earthquake?

A Try to save someone else's life as well as your own
B Make sure that no heavy weight have fallen on top of you.
C Try to get out of the building as quickly as possible.
D Stay inside and wait until the earthquake has finished

11 تفاصيل

According to paragraph (5), what is TRUE about a 'fight or flight' situation?

A All body functions are considered essential.
B All body functions are considered NON essential.
C The digestive and immune systems are considered NON essential.
D The digestive and immune systems are considered essential

12 مرجع الضمير

The word it in paragraph (4) refers to _____ "

A body B cortisol C adrenaline D noradrenaline

13 مرجع الضمير

Why does the author use the pronoun "we" throughout the passage?

A Because he is describing the experience of someone else.
B Because he distances himself from the experience he describes
C Because he is describing an experience unique to himself
D Because he believes that his feeling will be shared by many others

14 معنى كلمة

The word overloaded in paragraph (1), is closest in meaning to _____

A Feeling NOT listened to B Feeling NOT supported
C Feeling under pressure D Feeling overweight

04 الإجابات مع الشرح

9 D The ways a person's body reacts to a dangerous situation

السبب العبارة تصف استجابة الجسم للخطر: يجهز نفسه إما للهرب أو للمواجهة.

our body activates resources to protect us ... This is known as our 'fight-or-flight' response.

ليه الباقي غلط A و B كل واحد منهما نصف الاستجابة فقط (المواجهة أو الهرب)، والعبارة تعني ردة فعل الجسم كلها.

10 **C** Try to get out of the building as quickly as possible.

السبب الفقرة 3: كلما خرجت أسرع زادت فرصة النجاة.

> *the faster you can get yourself, and anyone else who might be at home, out the more likely you are all to survive*

11 **C** The digestive and immune systems are considered NON essential.

السبب الفقرة 5 تسمي الهضم والمناعة وظائف غير أساسية تتباطأ وقت الخطر.

> *Non-essential body functions slow down, such as our digestive and immune systems*

12 **A** body

السبب الجسم ينتج المواد الكيميائية، وهذه المواد تدفع «الجسم» لرفع نبض القلب؛ it مفرد ويعود على body.

> *our body produces larger quantities of the chemicals cortisol, adrenaline and noradrenaline, which trigger it to have a higher heart rate*

ليه الباقي غلط المواد الثلاث هي الفاعل which trigger، والضمير هو المفعول، فلا يعود عليها.

13 **D** Because he believes that his feeling will be shared by many others

السبب الكاتب يستخدم we لأنه يصف تجربة يمر بها أغلب الناس.

> *We generally use the word "stress" when we feel that everything seems to have become too much.*

14 **C** Feeling under pressure

السبب overloaded = محمّل فوق طاقته، والجملة بعدها تشرحها: نتساءل هل نقدر على المطالب = تحت ضغط.

> *We are overloaded and wonder whether we really can cope with the demands placed upon us.*

ليه الباقي غلط overweight فخ شكلي (زيادة الوزن)، ولا حديث عن عدم الاستماع أو الدعم.

05 المشاعر
Emotions

ذكرها المختبرون 108 مرة
5 سؤال

Emotions

(1) What comes to mind when you think of emotions? It's probably happiness, sadness, or fear. Emotions such as happiness, anger, sadness, and fear are important emotions, but human emotions experience is vast, which means that people are capable of experiencing a wide variety of feelings.

(2) Let's consider what emotions do and when emotions happen. According to functionalist theories of emotion, emotions help people manage important tasks. Fear, for example, prepares the body to fight or flee; happiness rewards achieving goals and builds attachments to other people. But what causes emotions to happen in the first place? Although it usually feels like something in the world causes an emotion, such as a good hug, emotion theories argue that emotions most likely come from how we think about what is happening in the world, not what is really happening. After all, if things in the world directly caused emotions, everyone would always have the same emotion in response to something. Interest, for example, may motivate a person to learn something new, whereas it may not create a response in someone else.

(3) For instance, appraisal theories propose that each emotion is caused by a group of appraisals, which are evaluations and judgments of what events in the world mean for our goals and well-being. We ask ourselves, is this relevant to me? Does it further or obstruct my goals? Can I deal with it or do something about it? Did someone do it on purpose? Different emotions come from different answers to these appraisal questions.

(4) In general, emotions play important roles in everyday challenges such as responding to threats and building relationships. But emotions also help in other, more intellectual challenges for humans. Compared with other animals, we are born with little knowledge but have the potential for enormous intelligence. Emotions such as surprise, interest, and confusion first signal that something out of the ordinary has happened that deserves our attention. They then motivate us to engage with the new things that strain our understanding of the world and how it works. Emotions surely help us to fight or flee, but I believe that, more importantly, they aid in learning, exploring, and reflecting.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

15 How are functionalist theories of emotions different from appraisal theories? — تفاصيل

A According to functionalist theories, emotions come from how we judge what happens in the world.
B According to functionalist theories, emotions help us build relationships or respond to threats.
C According to functionalist theories, events in the world cause emotions.
D According to functionalist theories, emotional intelligence is more important than logical intelligence.

16 What idea does the word "relationships" in Paragraph (4) repeat? — تفاصيل

A human emotional experience
B evaluations and judgments
C attachments to other people
D more intellectual challenges

17 What can we understand from Paragraph (2) about a good hug? — استنتاج

A It is likely the result of the two people involved feeling happy.
B Scientists argue it works as a reward for achieving a goal.
C It may cause different feelings in different people.
D Theories disagree over its emotional value.

18 Which of the following is the writer's opinion? — استنتاج

A People experience many emotions.
B People know very little at birth.
C Happiness and sadness are emotions.
D Emotions help us learn, explore, and reflect.

19 What can we understand from the passage about emotions? — استنتاج

A Emotions may sometimes prevent us from achieving our goals.
B Emotions make us want to know more about the world around us.
C Scientists have decided there are seven basic types of emotions.
D The latest research has changed the way we now think about emotions.

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05 الإجابات مع الشرح

15 **B** **According to functionalist theories, emotions help us build relationships or respond to threats.**

السبب النظريات الوظيفية ترى أن المشاعر تساعدنا في مهام مهمة مثل مواجهة الخطر وبناء العلاقات، أما نظريات التقييم فتقول إن المشاعر تأتي من حكمنا على الأحداث.

> *emotions play important roles in everyday challenges such as responding to threats and building relationships*

ليه الباقي غلط A هو رأي نظريات التقييم وليس الوظيفية، و C ترفضه القطعة، و D غير مذكور.

16 **C** **attachments to other people**

السبب كلمة relationships تكرر فكرة attachments to other people في الفقرة 2.

> *happiness rewards achieving goals and builds attachments to other people*

17 **C** **It may cause different feelings in different people.**

السبب القطعة تقول إن المشاعر تأتي من طريقة تفكيرنا في الحدث، فنفس الحدث (مثل الحضن) قد يعطي مشاعر مختلفة.

> *if things in the world directly caused emotions, everyone would always have the same emotion in response to something*

18 **D** **Emotions help us learn, explore, and reflect.**

السبب الرأي يبدأ بعبارة I believe في آخر القطعة؛ باقي الخيارات حقائق.

> *but I believe that, more importantly, they aid in learning, exploring, and reflecting.*

19 **B** **Emotions make us want to know more about the world around us.**

السبب مشاعر الدهشة والاهتمام تدفعنا للتعامل مع الأشياء الجديدة وفهم العالم.

> *They then motivate us to engage with the new things that strain our understanding of the world*

ستيب STEP تجميعات متكررة

06 استبيان الوجبات السريعة (جدول)
Fast Food Survey (table)

ذكرها المختبرون 106 مرة
2 سؤال

How many fast food meals do you eat per week?

	Children 5-12	Young Adults 13-17	Adults 18-49	Elderly 50+	Total
0	8	2	4	9	23
1-3	24	4	26	31	85
4-6	10	28	14	6	58
7+	8	16	6	4	34

Which fast food restaurants have you visited in the last year?

	Children 5-12	Young Adults 13-17	Adults 18-49	Elderly 50+	Total
McDonalds	44	42	41	42	169
Burger King	36	41	32	28	137
Herfy	11	43	32	3	89
Hardees	9	46	25	10	90
KFC	24	36	43	42	145

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

20 Which fast food restaurant is most popular for adults? — تفاصيل

A Burger King B McDonalds C Hardees D KFC

21 Which fast food restaurant is most popular? — تفاصيل

A McDonalds B Hardees C Herfy D KFC

06 الإجابات مع الشرح

20 D KFC

السبب: في عمود Adults 18-49 أعلى رقم هو 43 = KFC، ثم 41 = McDonalds.

KFC: 24 / 36 / 43 / 42 / 145

ليه الباقي غلط: انتبه: السؤال عن عمود البالغين فقط، وليس عن المجموع.

21 **A McDonalds**

السبب الأكثر شعبية = أعلى رقم في عمود Total، وهو 169 = McDonalds.

McDonalds: 44 / 42 / 41 / 42 / 169

ستيب STEP: تجميعات متكررة

07 الرهاب Phobias

ذكرها المختبرون 105 مرة

2 سؤال

Passage A: Phobias

(1) A phobia is a specific fear of a certain object, situation, or activity. Most people learn to live with their phobias, but for others the fear can be so debilitating that they go to extremes to avoid the scary situation. A person who suffers from arachnophobia (fear of spiders), for example, may refuse to enter a room until it has been checked for spiders, or may refuse to take a vacation in the countryside because spiders may be there.

(2) A common phobia is social phobia, extreme shyness around people or anxiety in social situations. Social phobia may be specific to a certain event, such as using a public restroom, or it can be more generalized anxiety toward almost all people outside of close family and friends. People with social phobia will often experience physical symptoms in public, such as sweating, blushing and nausea. They are convinced that everybody around them notices these symptoms as they are occurring.

(3) The most incapacitating phobia is agoraphobia, defined as anxiety about being in places from which escape might be difficult. Typical places that provoke the panic attacks are parking lots: crowded streets or shops; and bridges, tunnels, or expressways. People (mostly women) who suffer from agoraphobia may have great difficulty leaving their homes and interacting with other people.

(4) Phobias affect about 9% of American adults, and they are about twice as prevalent in women as in men. In most cases phobias first appear in childhood and adolescence, and usually persist into adulthood.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

تفاصيل

22 **What do passages A and B say about the development of phobia?**

A It can result in over-dependence on drugs.

B It can lead to the sufferer NOT leaving home.

C It starts in childhood but disappears in adulthood.

D It usually persists across different stages of human life.

ستيب STEP: تجميعات متكررة

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23 **According to passages A and B, what is one defining feature of social phobia?** تفاصيل

- **A** Fear of close family member.
- **B** Intense anxiety when dealing with people.
- **C** Violent body movements in social situations.
- **D** Feeling anxious about being in crowded streets.

07 الإجابات مع الشرح

22 **D It usually persists across different stages of human life.**

السبب الرهاب يظهر غالبًا في الطفولة والمراهقة ويستمر إلى سن البلوغ، أي عبر مراحل العمر.

> *In most cases phobias first appear in childhood and adolescence, and usually persist into adulthood.*

ليه الباقي غلط C عكس القطعة: لا يختفي في الكبر. و B صحيح لنوع واحد فقط (رهاب الأماكن) وليس عن تطوّر الرهاب.

23 **B Intense anxiety when dealing with people.**

السبب الرهاب الاجتماعي = خجل شديد أو قلق عند التعامل مع الناس.

> *A common phobia is social phobia, extreme shyness around people or anxiety in social situations.*

ليه الباقي غلط D (الشوارع المزدحمة) هو رهاب الأماكن agoraphobia، والقلق يكون مع الناس «خارج» العائلة والأصدقاء.

ستيب STEP: تجميعات متكررة

ذكرها المختبرون 100 مرة

5 سؤال

08 علاج السرطان
Conquering Cancer

(1) Cancer is a serious disease, but when a tumor is localized, it can be removed by surgery. Skin cancer, the most common form of the disease, can usually be treated this way. Melanomas, the most serious form of skin cancer, can be removed surgically, but only if spotted very early.

(2) Other forms of treatment make use of the fact that cancer cells grow rapidly and, therefore, need to copy their DNA more quickly than do most normal cells. This makes them especially vulnerable to damage from radiation. As a result, many tumors can be effectively treated with carefully targeted beams of radiation.

(3) Medical researchers have worked for years to develop chemical compounds that would kill cancer cells, or at least slow their growth. The use of such compounds against cancer is known as chemotherapy. Great advances in chemotherapy have taken place in recent years and have even made it possible to cure some forms of cancer. However, because most chemotherapy compounds target rapidly dividing cells, they also interfere with cell division in normal, healthy cells. This produces serious side effects in many patients, and it is one of the reasons why scientists are so interested in gaining a better understanding of cell cycle proteins in cancer. The goal of many researchers is to find highly specific ways in which cancer cells can be targeted for destruction while leaving healthy cells unaffected.

(4) Understanding and combating cancer remains a major scientific challenge, but scientists at least know where to start. Cancer is a disease of the cell cycle, and conquering cancer will require a much deeper understanding of the processes that control cell division.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

24 What does paragraph (1) say about skin cancer?

تفاصيل

A It's a less frequent than other forms of cancer.
B It can normally be removed by surgery.
C People can get it from sunbathing.
D Melanomas are easy to notice.

25 How can scientists avoid destroying healthy cells?

تفاصيل

A by treating cancer very early.
B by copying the DNA in cancer cells.
C by using chemotherapy or beams of radiation.
D by better understanding cell cycle proteins in cancer.

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26 معنى كلمة

Which word or words can we use to replace the word "therefore" in paragraph (2)?

A instead B besides C in addition D as a result

27 استنتاج

Why does the writer use the word "However" in paragraph (3)?

A to say that chemotherapy has its dangers
B to give an example of how chemotherapy works
C to repeat the previous sentence with fewer words
D to say that the previous sentence is actually incorrect.

28 استنتاج

How does the writer feel about our understanding of cancer?

A He feels hopeful because scientists have made at least some progress.
B He feels excited because scientists are close to curing its most common forms.
C He feels angry because some treatments make people feel ill.
D He feels disappointed because few of the treatments work.

08 الإجابات مع الشرح

24 B It can normally be removed by surgery.

السبب: سرطان الجلد يمكن غالبًا علاجه بالجراحة، و normally = usually.

Skin cancer, the most common form of the disease, can usually be treated this way.

ليه الباقي غلط: A عكس القطعة (هو الأكثر شيوعًا)، و D غلط لأن الميلانوما تحتاج اكتشافًا مبكرًا جدًا.

25 D by better understanding cell cycle proteins in cancer.

السبب: العلماء يريدون فهم بروتينات دورة الخلية ليستهدفوا الخلايا السرطانية ويتركوا السليمة.

scientists are so interested in gaining a better understanding of cell cycle proteins in cancer

ليه الباقي غلط: C غلط: العلاج الكيميائي هو اللي يضر الخلايا السليمة.

26 D as a result

السبب: therefore = لذلك، وتساويها as a result (وهي مستخدمة في نفس الفقرة).

cancer cells grow rapidly and, therefore, need to copy their DNA more quickly

ليه الباقي غلط: besides و in addition للإضافة، و instead للبديل.

27 A to say that chemotherapy has its dangers

السبب: بعد However يذكر الكاتب أضرار العلاج الكيميائي على الخلايا السليمة، فهي تقلب الكلام من المدح إلى الخطر.

However, because most chemotherapy compounds target rapidly dividing cells, they also interfere with cell division in normal, healthy cells.

28 **A** He feels hopeful because scientists have made at least some progress.

السبب الكاتب يرى أن السرطان تحدٍّ كبير لكن العلماء يعرفون من أين يبدؤون، وهذا تفاؤل.

> *Understanding and combating cancer remains a major scientific challenge, but scientists at least know where to start.*

ستيب STEP: تجميعات متكررة

09 النحل Bees

ذكرها المختبرون 100 مرة

5 سؤال

(1) The Holy Quran tells us that insects form communities (Surat al-An'am, 38) and that they can communicate with one another (Surat an-Naml, 18) and work together to produce food (Surat al-Nahl, 68-9). Modern science, more than 1,400 years afterwards, has accepted that this information is indeed correct. Now, scientists in the United Kingdom are learning that some of these amazing creatures may also have feelings and emotions similar to those of humans.

(2) At Newcastle University, they are trying to discover how some insects feel. In recent years, honeybee numbers have been declining worldwide at a disturbing rate. Many explanations have been given for this. Some say the bees are dying because of disease and pesticides. Others say it may be due to pollution or microwave towers. At any rate, this is a very serious situation, for humans as well as for the bees. At least one-third of the world's crops depend on bees. Farmers are worried. What about the bees? How do they feel? Are they scared, sad or depressed?

(3) The researchers trained a group of bees to tell the difference between two smells. After one smell, they were always given a sweet sugar reward. After the other smell, they were given a bitter substance. They quickly learned to choose the smell with the sweet prize and to stick out their tongues to get it. The scientists wanted to see what the bees would do if they were put in a stressful situation. They shook up half of the bees; to make them think they were in danger. These stressed bees refused to put out their tongues for new smells, and only chose the old smell which they knew would be followed by a sweet treat. Even though there was an equal chance that a strange smell would taste good, it seemed they believed it would taste bad and were afraid to try it. These bees had become “pessimists". The other half-the unstressed bees-were much more likely to try tastes from new smells, thus remaining “optimists”.

(4) When humans are stressed, anxious, worried or depressed, they have lower levels of the brain hormone serotonin. So did the stressed bees. This may mean that the feelings of anxious and stressed bees are similar to those of humans with mood disorders. This knowledge, along with further investigations into bee behavior and "feelings" can hopefully be used to understand and remedy their declining numbers, for their sake and for ours.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

29 What is the main idea of Paragraph (4)? — الفكرة الرئيسية

A Humans and bees have the same hormones.

B Humans are more affected by stress than bees.

C Understanding similar feelings in humans and bees may help both.

D Training stressed bees is important for treating human mood disorders.

30 Why are bees important to humans? تفاصيل

A They have communities and can communicate.
B They are necessary to produce our food crops.
C They are becoming more numerous than us.
D They have feelings like us.

31 Which paragraph describes how the scientists trained the bees? تفاصيل

A Paragraph (1) B Paragraph (2) C Paragraph (3) D Paragraph (4)

32 The words these amazing creatures in Paragraph (1) refer to _____ مرجع الضمير

A bees B insects C scientists D communities

33 What can we understand about optimists in Paragraph (3)? استنتاج

A They are afraid to try new experiences.
B They prefer the smells they learn first.
C They are NOT afraid to try new things.
D They feel stressed and worried.

09 الإجابات مع الشرح

29 C Understanding similar feelings in humans and bees may help both.

السبب الفقرة 4 تنتهي بأن فهم مشاعر النحل قد يساعد النحل والإنسان معًا.

can hopefully be used to understand and remedy their declining numbers, for their sake and for ours

30 B They are necessary to produce our food crops.

السبب ثلث محاصيل العالم على الأقل يعتمد على النحل.

At least one-third of the world's crops depend on bees.

31 C Paragraph (3)

السبب الفقرة 3 تبدأ بتدريب النحل على التمييز بين رائحتين.

The researchers trained a group of bees to tell the difference between two smells.

ستيب STEP: تجميعات متكررة

32 **B** insects

السبب الجملة الأولى عن الحشرات insects، و these amazing creatures ترجع لها؛ النحل لم يُذكر بعد في الفقرة 1.

> *The Holy Quran tells us that insects form communities*

33 **C** They are NOT afraid to try new things.

السبب النحل غير المضغوط كان يجرب الروائح الجديدة، فسُمّي متفائلًا: لا يخاف من التجربة.

> *The other half-the unstressed bees-were much more likely to try tastes from new smells, thus remaining "optimists"*

ليه الباقي غلط A و B و D صفات النحل المتشائم المضغوط.

ستيب STEP: تجميعات متكررة

10

العدوانية عند الأطفال
Aggression in Children

ذكرها المختبرون 99 مرة

4 سؤال

(1) Most parents do not enjoy watching their children misbehave. Bad behavior by children often results in bad behavior by parents. The first idea that has to go is that poor parenting is what makes children misbehave or behave badly. The cause of the child's behavior may lie in psychological problems; it may even lie in the biology of the child. Whatever the reason, the important issue is how parents respond when children act out.

(2) Aggressive behavior starts early in children's lives. A study looked at children 6 to 24 months of age. Mothers reported how often their children acted out, or displayed poor behavior. They cited kicking, hitting, pulling hair, biting and even hurting animals. The most common actions were hitting or smacking another person. The least common action was hurting animals. Hitting someone peaked at 18 months when eight of 10 kids were hitting and smacking other people. Kicking, pushing and throwing things peaked at 20 months. The "terrible twos" started before the age two.

(3) Pediatricians can assure parents that these behaviors are normal in small children. The big job is to guide parents on how to deal with their children when they behave this way. From the start, parents should set limits. They need to respond in ways that may help redirect or distract a child. This would help a child go from misbehaving to a more positive behavior. Pediatricians believe these approaches work. It seems certain that positive responses such as these are better than anger or punishment.

(4) There are some questions that parents must also consider. Does the child have any issues that he or she is dealing with? Does the child have friends? Some children may show signs of depression when they have had a negative experience. The answers to these questions may explain why the child is acting out.

(5) Parents often use their instincts in dealing with their children without considering what is causing the misbehavior. They may respond with anger, threats, and punishment. A better way is to plan in order to avoid potential conflicts. For example, if getting ready for school is a problem, parents can come up with ways to make it easier. This would help them to avoid potential problems before they start.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

34 What word other than "misbehave" does the writer use for aggressive behavior?

معنى كلمة

A act out
B avoid
C distract
D deal with

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35 استنتاج

Which of the following is a fact?

A Poor parenting is the cause of children's misbehavior.
B Aggressive behavior is normal in young children.
C Parents often deal with children's bad behavior well.
D Hurting animals is the most common aggressive behavior among children.

36 استنتاج

How is the writer's view about children's bad behavior different from the parents' view?

A The writer believes that parents should plan to avoid bad behavior.
B The writer believes that parents should use their instincts to deal with bad behavior.
C The writer believes that parents should respond to bad behavior with threats and punishment.
D The writer believes that parents should ignore bad behavior.

37 استنتاج

What can we understand about parents in Paragraph (5)?

A Parents fail to prevent children's bad behavior.
B Parents blame their children's friends for bad behavior.
C Parents consider the causes of children's bad behavior.
D Parents think schools should deal with children's bad behavior.

10 الإجابات مع الشرح

34 A act out

السبب: الكاتب يستخدم act out بمعنى السلوك السيئ، وتفسيرها في الفقرة 2.

Mothers reported how often their children acted out, or displayed poor behavior.

35 B Aggressive behavior is normal in young children.

السبب: حقيقة يؤكدها أطباء الأطفال: هذه السلوكيات طبيعية عند الصغار.

Pediatricians can assure parents that these behaviors are normal in small children.

ليه الباقي غلط: A ترفضه القطعة صراحة، و D عكسها: إيذاء الحيوانات هو الأقل.

36 A The writer believes that parents should plan to avoid bad behavior.

السبب: الآباء يتصرفون بالغريزة (غضب وعقاب)، والكاتب يرى أن الأفضل التخطيط لتجنب المشكلة.

A better way is to plan in order to avoid potential conflicts.

ليه الباقي غلط: B و C هما تصرف الآباء الذي ينتقده الكاتب.

ستيب STEP: تجميعات متكررة

37 **A** Parents fail to prevent children's bad behavior.

السبب الآباء يتصرفون بعد وقوع السلوك بالغضب والعقاب، ولا يخططون لمنعه قبل أن يبدأ.

> *Parents often use their instincts in dealing with their children without considering what is causing the misbehavior.*

ليه الباقي غلط C عكس القطعة: هم لا يفكرون في السبب.

ستيب STEP: تجميعات متكررة

11

السمنة
Obesity (Passages A and B)

ذكرها المختبرون 95 مرة

3 سؤال

Passage A: Environmental Causes of Obesity

(1) Genetic factors are the forces inside you that cause you to become and stay overweight. On the other hand, environmental factors are the outside forces that contribute to these problems. They include anything in our environment that makes us more likely to eat too much or exercise too little. Taken together, experts think that environmental factors are the driving force for the causes of obesity and its dramatic rise.

(2) Environmental influences come into play very early in life, even before we are born. Researchers believe that the environment surrounding babies before they are born determines how they will develop later on in life. For example, babies of mothers who smoked during pregnancy are more likely to become overweight or develop a disease than those whose mothers did not smoke.

(3) Childhood habits often stick with people for the rest of their lives. Kids who drink sodas and other drinks that contain sugar and eat high-calorie foods develop a taste for these products and continue eating them as adults. These behaviors tend to promote weight gain. Similarly, kids who watch television and play video games instead of being active may be programming themselves for a sedentary future, which involves sitting down a lot rather than moving around.

(4) In short, today's environment promotes obesity by encouraging us to eat more and exercise less. There is also growing evidence that broader aspects of the way we live – such as how much we sleep, our stress levels, and other psychological factors – can affect weight as well.

Passage B: The Food Factor in Obesity

(1) Today, Americans are eating more calories on average than they did in the 1970s. Experts believe that what is driving this trend is a combination of the increased availability and bigger portions of higher calorie foods.

(2) Food is readily available practically everywhere we go-shopping malls, stadiums, movie theaters. Today more Americans are eating outside the home. For example, Americans spent 27% of their food budget on food they ate outside the home; by 2006, that percentage had risen to 46%.

(3) In the 1950s, fast-food restaurants offered one portion size. Today, portion sizes have suddenly increased, a trend that has spread in many other types of food, from cookies to sandwiches. For example, a single "supersized" meal may contain 1,500-2,000 calories, which are equal to all the calories that most people need for an entire day.

(4) This is important because research shows that people will often eat what is in front of them, even if they are already full. We are also eating more high-calorie foods (especially salty snacks, soft drinks, and pizza), which are much more readily available than lower calorie choices, like salads and whole fruits. Research also shows that what children eat when they are younger will affect what they will eat as they grow older. Children who eat fast food are likely to continue to eat

this type of food later on in life, which would cause them many health problems. Doctors are now encouraging parents to help their children to develop healthy eating habits from a younger age in order to avoid future health complications.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

38 What do passages A and B say about the cause of obesity? — تفاصيل

A Bad eating habits cause obesity
B Lack of exercise causes obesity.
C Stress factors cause obesity.
D Genetics cause obesity

39 What do passages A and B say about today's environment? — تفاصيل

A People exercise less.
B People are healthier.
C People eat a lot more.
D People have more stress.

40 What can we understand from passages A and B about children's eating habits? — استنتاج

A They are less healthy than they were in the past.
B They affect their eating habits as adults.
C They are linked to levels of activity.
D They usually improve as they get older.

11 الإجابات مع الشرح

38 A Bad eating habits cause obesity

السبب: القطعتان تتكلمان عن الأكل: عادات الطفولة (المشروبات السكرية) في A، والحصص الكبيرة والأكل خارج البيت في B.

today's environment promotes obesity by encouraging us to eat more and exercise less

ليه الباقي غلط: قلة الرياضة والتوتر مذكورة في القطعة A فقط، والجينات ليست موضوع القطعتين.

39 C People eat a lot more.

السبب: الفكرة المشتركة في القطعتين: الناس يأكلون أكثر من قبل.

Today, Americans are eating more calories on average than they did in the 1970s.

ليه الباقي غلط: A مذكور في القطعة A فقط، والسؤال عن القطعتين معًا.

40 B They affect their eating habits as adults.

السبب: القطعتان تقولان إن ما يأكله الطفل يستمر معه في الكبر.

Research also shows that what children eat when they are younger will affect what they will eat as they grow older.

ستيب STEP: تجميعات متكررة

12 مدخل إلى علم النفس
Introducing Psychology

ذكرها المختبرون 79 مرة
4 سؤال

(1) Psychology is the scientific study of mind and behavior. The word “psychology” comes from the Greek words “psyche, meaning life, and “logos,” meaning explanation. Psychology is a popular major for students, a popular topic in the public media, and a part of our everyday lives. Television shows often feature psychologists who provide personal advice to those with personal or family difficulties. Crime dramas such as CSI, Lie to Me, and others feature the work of forensic psychologists who use psychological principles to help solve crimes. And many people have direct knowledge about psychology because they have visited psychologists, for instance, school counselors, family therapists, and religious, marriage, or bereavement counselors.

(2) Because we are frequently exposed to the work of psychologists in our everyday lives, we all have an idea about what psychology is and what psychologists do. In many ways, your conceptions are probably correct. Psychologists do work in forensic fields, and they do provide counseling and therapy for people in distress. But there are hundreds of thousands of psychologists in the world, and most of them work in other places, doing work that you are probably now aware of.

(3) Most psychologists work in research laboratories, hospitals, and other field settings where they study the behavior of humans and animals. For instance, experts in the Psychology Department at the University of Maryland study such diverse topics as anxiety in children, the interpretation of dreams, the effects of caffeine on thinking, how birds recognize each other, how people from different cultures react differently in negotiation. Psychologists also work in schools and businesses, and they use a variety of methods, including observation, questionnaires, interviews, and laboratory studies, to help them understand behavior.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

41 What do psychologists do to help people who are unhappy in their lives?
تفاصيل

A They advise them to stop drinking coffee.
B They provide them with treatment services.
C They take them to special hospitals for help.
D They provide them with advice about their food.

42 What does the passage say about the work of most psychologists
تفاصيل

A It is field research.
B It is forensic science.
C It is TV entertainment.
D It is personal counseling.

معنى كلمة

43 **Which word can we use to replace the word "But" in paragraph (2)?**

A In addition B Therefore C Finally D However

استنتاج

44 **What does the writer think about our knowledge of psychology?**

A It is complete. B It is INCORRECT.
C It is from university. D It is from experience.

12 الإجابات مع الشرح

41 **B They provide them with treatment services.**

السبب: الأخصائيون النفسيون يقدمون الإرشاد والعلاج لمن هم في ضيق.

they do provide counseling and therapy for people in distress

42 **A It is field research.**

السبب: أغلب الأخصائيين يعملون في المختبرات والمستشفيات و «الميدان» يدرسون السلوك = بحث ميداني.

Most psychologists work in research laboratories, hospitals, and other field settings

ليه الباقي غلط: الطب الجنائي والإرشاد والبرامج التلفزيونية هي الصورة الشائعة، والكاتب يقول إن أغلبهم يعملون في أماكن أخرى.

43 **D However**

السبب: But تربط بين فكرتين متعاكستين، ومثلها However.

But there are hundreds of thousands of psychologists in the world

44 **D It is from experience.**

السبب: معرفتنا بعلم النفس تأتي من تعاملنا اليومي مع عمل الأخصائيين.

Because we are frequently exposed to the work of psychologists in our everyday lives, we all have an idea about what psychology is

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13

الحاسب
Computers (Passages A and B)

ذكرها المختبرون 79 مرة

3 سؤال

Passage A: What is a computer?

All computer systems, no matter which size, can be divided into four basic elements that perform certain tasks. 1-INPUT – Methods of getting instructions and data into the computer. 2-PROCESS – the central processing unit (CPU) processes data according to instructions. Actions include arithmetic operations, comparison, data transfer. Main memory is used to store temporary data and software in current use, and is also involved in running the software. 3-STORE – Ways to file data for future use onto what is known as 'backing store'. 4-OUTPUT – Methods to produce information that is going out.

A computer system is a series of devices which are capable of doing 1-4 above. Altogether this equipment is called hardware. Devices other than the computer itself are called peripheral (external devices). Computers carry out instructions – a set of which is called a program which will be written in a computer language. A program controls the computer by telling the computer's control unit what to do. Software is the general term to describe all the programs and data used in a computer system. Interface is the hardware and linked software needed between processors and peripherals to make up for the differences in their operating speeds and codes.

Passage B: CPU and Memory

The performance of the CPU is affected by the number of cores, clock speed and memory. There are different types of volatile and non-volatile memory.

Computers work with data, and the equipment enabling this is called hardware. Computers require input hardware, processing hardware and output hardware. The hardware that defines a computer is the CPU and memory. Without these a computer could not function.

The CPU and memory work together to run programs. The CPU executes programs using the fetch-decode-execute cycle. The memory stores program operations and data while a program is being executed. There are several types of memory, including: registers, cache, RAM and virtual memory. The storage stores programs and file long term, even when they are not in use. Devices such as hard drives, USB memory sticks or SD cards are used to store files such as photos, music and software applications long term.

Most computers have a CPU and a separate graphics processing unit (GPU) for images. The (GPU) can be.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

45 What do passages A and B say about the CPU? — تفاصيل

- **A** It carries out arithmetic operations and controls internet access.
- **B** It is a piece of hardware that works together with memory.
- **C** It helps transfer data and functions as virtual memory.
- **D** It controls the computer's clock and the GPU

46 What can we understand from passages A and B about memory? — استنتاج

- **A** It allows a computer to perform its basic functions and affects CPU performance.
- **B** It is important for memory and the CPU to be capable of the same speeds.
- **C** It helps run programs and works as a temporary data storage.
- **D** It can be expanded and be volatile or non-volatile.

47 What can we understand from passages A and B about hardware? — استنتاج

- **A** It enables data input, data output and data processing.
- **B** It includes the CPU, memory and power supply.
- **C** It is a type of peripheral, or external device.
- **D** It controls the CPU's operating speed.

13 الإجابات مع الشرح

45 **B It is a piece of hardware that works together with memory.**

السبب: القطعة B تقول إن المعالج والذاكرة يعملان معًا لتشغيل البرامج، والقطعتان تعدّانه من العتاد.

The CPU and memory work together to run programs.

46 **C It helps run programs and works as a temporary data storage.**

السبب: الذاكرة تشارك في تشغيل البرامج وتخزن البيانات مؤقتًا، وهذه الفكرة في القطعتين.

Main memory is used to store temporary data and software in current use, and is also involved in running the software.

ليه الباقي غلط: A و D أفكار من القطعة B وحدها، و B غير مذكور.

47 **A It enables data input, data output and data processing.**

السبب: الحاسب يحتاج عتادًا للإدخال وللمعالجة وللإخراج.

Computers require input hardware, processing hardware and output hardware.

ستيب STEP: تجميعات متكررة

14 النفايات السامة
Toxic Waste

ذكرها المختبرون 75 مرة
3 سؤال

(1) The unwanted poisonous by-products of human activity, toxic wastes can arise from many sources. Atmospheric pollution, for example, is caused by automobiles, power plants, and incinerators. Acid rain is produced when oxides of sulfur, carbon, and nitrogen from burning fossil fuels combine with moisture in the air, producing hazardous acids. Discarded industrial solvents include low-molecular weight material such as benzene and carbon tetrachloride and high-molecular weight organic materials such as PCBs (polychlorinated biphenyls). which were used in the past for hydraulic fluids and in electrical transformers and capacitors. Dioxins are the unwanted by-products of the manufacture of herbicides. Heavy-metal residues from copper or silver mining can contaminate groundwater or streams and lakes and become concentrated in fish. Radioactive wastes are generated by nuclear power reactors, the manufacture of nuclear weapons, and, at low levels, by the industrial and medical application of radioisotopes.

(2) In 1981 approximately 264 million tons of waste were produced in the United States — about 5.5 million tons were considered hazardous. During 1986 the United States generated about 1,770,000 cubic feet (50,000 cubic meters) of low-level solid and liquid radioactive waste.

(3) Many toxic wastes can harm humans, animals, or plants. Significant exposure to lead in paints and gasoline can lead to nervous disorders. Organic toxic wastes can lead to liver and kidney inflammation. Others, such as PCBs, are suspected of causing birth defects. Some organic materials, such as benzene, are known to cause cancer in laboratory animals and are listed as possible cancer producing agents in humans. Moderately high levels of copper, silver, and other heavy metals are detrimental to the growth of algae.

(4) Some toxic wastes cause ecological imbalances. The insecticide DDT (dichloro-diphenyl trichloroethane), for example, almost wiped out the American bald eagle population and reduced many other bird populations. It destroyed the birds' ability to develop a sufficiently strong egg shell to assure hatching. Because of DDT's threat to bird populations severe restrictions were imposed in 1972 on its use in the United States. Other laws required the reduction of lead in gasoline and the termination of PCB production. DDT is still used in developing countries for its effectiveness against the malaria-carrying mosquito.

(5) Depending on the type of waste, toxic wastes can be treated chemically, biologically, or by combustion. (See also garbage and refuse disposal.) An acid spill, for example, may be treated with lime or bicarbonate to neutralize the acid. Oil spills at sea are often treated with an emulsifier to disperse the oil in the water, thereby reducing the adverse effects on beaches and on wildlife. Effective methods of biological treatment of toxic wastes are still undergoing development. Organisms that can survive the toxicity and degrade the toxic wastes in a reasonable amount of time have not been found, but advances in genetic engineering may change this situation.

(6) Combustion and other high-temperature treatments are the major methods used to reduce the toxicity of wastes. Processes involving plasma technology are used for specific toxic wastes. These processes use high temperatures to ionize a mixture of air and fuel gases. When combined

with these ionized gases, hot toxic gases from the waste are decomposed into relatively harmless components. Plasma treatment at temperatures above 1,650° F (900° C) can diminish the toxicity of gas emissions such as dioxin vapor by about ten times the reduction that results from conventional combustion treatments.

(7) Incineration of liquid wastes is the primary method of toxic waste treatment in the United States. It usually involves injecting the waste as small droplets into a burner. A fixed-hearth process, by which both liquid and solid wastes can be handled, is also used. The waste is treated in two stages. It is first partially burned with insufficient air and then burned again with excess air at higher temperatures to destroy the gases released during the first stage.

(8) Another widely used process involves incineration in a rotary kiln. This is used for solids, liquids, their mixtures (slurries), and wastes in containers. Some hazardous wastes are disposed of directly as fuel in industrial boilers or cement kilns.

(9) Liquid toxic wastes must be stored temporarily before disposal. Steel drums are used for short-term storage. Materials less subject to corrosion, such as glass or ceramic, are often required for long-term storage. Ocean dumping and the use of landfills are no longer permitted for the disposal of hazardous waste in the United States.

(10) In the United States high-level nuclear wastes containing plutonium or uranium are either sent to reprocessing plants or stored near the nuclear facility site until disposal. Vitrification, a disposal procedure developed in recent years, is a process that embeds nuclear wastes in glasslike substances that are stable for thousands of years. These erosion-resistant blocks are then stored in underground salt mines or stable rock formations. Vitrification has been highly developed in France where nuclear energy is used extensively for electricity production. In the United States low-level radioactive wastes are currently stored at sites in three states — South Carolina, Nevada, and Washington — often after considerable volume reduction by compacting. Some low-level radioactive wastes are also incinerated. In the United States performance standards and permit requirements for thermal toxic waste destruction are determined by the Environmental Protection Agency (EPA). Current incinerator standards require the removal of at least 99.99 percent of the principal organic hazardous constituents and 99 percent of the hydrogen chloride in exhaust gases. Separate standards apply to solid particles (dust and ash) emitted during incineration. Radioactive wastes are regulated by the EPA, the Department of Energy (DOE), and the Nuclear Regulatory Commission (NRC). Each state is responsible for its own waste and interstate cooperation regarding disposal facilities is encouraged. Toxic waste regulation in Canada is monitored by the Ministry of the Environment, while the Atomic Energy Control Board manages radioactive waste.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

48 **According to the passage, what are the heavy metals that affect the growth of algae?** تفاصيل

A Copper and DDT
B Copper and silver
C Silver and benzene
D Gasoline and benzene

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49 What are three things needed for generating radioactive Wastes? تفاصيل

A nuclear power reactors, the manufacture of nuclear weapons, and the application of radioisotopes.
B the application of radioisotopes, the manufacture of herbicides and nuclear power reactors.
C the manufacture of nuclear weapons, acid rain and industrial boilers
D gas emissions, carbon tetrachloride and nuclear

50 What are three methods currently used for toxic waste treatment? تفاصيل

A combustion, ocean dumping and vitrification
B vitrification, landfills and biological treatment
C vitrification, combustion and chemical treatment
D combustion, biological treatment and vitrification

14 الإجابات مع الشرح

48 **B Copper and silver**

السبب: النحاس والفضة معادن ثقيلة تضر بنمو الطحالب.

Moderately high levels of copper, silver, and other heavy metals are detrimental to the growth of algae.

ليه الباقي غلط: DDT مبيد حشري، و benzene و gasoline ليست معادن.

49 **A nuclear power reactors, the manufacture of nuclear weapons, and the application of radioisotopes.**

السبب: الفقرة 1 تذكر مصادر النفايات المشعة الثلاثة بنفس الترتيب.

Radioactive wastes are generated by nuclear power reactors, the manufacture of nuclear weapons, and, at low levels, by the industrial and medical application of radioisotopes.

50 **C vitrification, combustion and chemical treatment**

السبب: المعالجة الكيميائية والحرق مستخدمة الآن، والتزجيج vitrification طريقة حديثة للنفايات النووية.

toxic wastes can be treated chemically, biologically, or by combustion ... Vitrification, a disposal procedure developed in recent years

ليه الباقي غلط: الإلقاء في المحيط والمدافن ممنوعة الآن، والمعالجة البيولوجية «ما زالت قيد التطوير».

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15

قنوات المياه الرومانية
Engineering Innovation (Roman Aqueducts)

ذكرها المختبرون 64 مرة

3 سؤال

(1) Rome is known for many things: its military conquests, its civic architecture, temples, roads, emperors, and sculpture. Yet none of these would have been possible without the most vital resource of all: water. Now, as then, water is life, and without effective distribution, there would have been no great Roman civilization. Even until relatively modern times, Roman techniques to collect, store, and channel water over huge distances remained unsurpassed.

(2) Such technology, was not, of course, invented from scratch by the Romans, and many earlier Mediterranean peoples had poured resources and expertise into managing water. On the island of Crete, the Minoans developed sophisticated rain-harvesting and filtering systems as early as the middle of the third millennium B.C. Cretan water management techniques were later adopted across the Greek-speaking world, and examples abound of tunnels, drainage systems, and cisterns, sometimes of considerable size.

(3) Yet although the water management tradition Rome inherited was rich and extensive, no previous system came close to the sophistication and reach of the Roman aqueduct. Striding across the landscape from Spain to Syria, these awe-inspiring structures not only carried life and livelihood but also proclaimed the greatness of Rome.

(4) Timeline: Bringing Water to Rome

312 B.C. The censor Appius Claudius Caecus builds Rome's first aqueduct, the Aqua Appia, which runs almost entirely underground.

144 B.C. Work begins on Rome's longest aqueduct, the 56-mile-long Aqua Marcia. The city has doubled in size since the last channel was built.

33 B.C. After the chaotic civil wars Octavian (later Emperor Augustus) improves Rome's water by building the Aqua Julia.

19 B.C. Marcus Agrippa, Augustus's son-in-law, oversees the building of the Aqua Virgo to supply the thermal baths in the Campus Martius.

A.D. 38-52 Caligula begins a new aqueduct to meet increased demand from baths. Claudius finishes the work and calls it Aqua Claudia.

A.D. 109 Trajan builds the Aqua Traiana, which brings water from near Lake Bracciano to supply Rome's new suburbs, known today as Trastevere.

(5) The Lifeline of Empire Aqueducts were costly public works, and not all Roman cities necessarily required them. Some cities, such as Pompeii, had their water needs met by wells or public and private cisterns dug beneath houses. Some cisterns could reach a colossal size, such as the Basilica Cistern (Yerebatan Sarnıcı) in Constantinople (now Istanbul, Turkey) and the Piscina Mirabilis in Miseno, Italy. The latter, built to provide drinking water to the Roman navy in the Bay of Naples, had a capacity of just under half a million cubic feet. Its colossal vault is held up by 48 pillars. Some cities needed much more water than cisterns could provide. Booming populations

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such as Rome's — thought to have reached one million in the first century A.D. — needed an entire system of aqueducts not only for drinking water but also for supplying ornamental public fountains and baths.

(6) The popular image many people have of an aqueduct is probably something like the spectacular bridge structure of the Pont du Gard in southern France. These aboveground arches were, in fact, only a small section of an aqueduct system. Roman engineers would create a gentle downward slope all the way from start to finish, since the only force powering the water's progress was gravity. Only valleys or gullies necessitated a monumental arched structure. For most of its route, water ran along underground or ground-level channels. Rome, for example, was supplied by aqueducts totaling 315 miles in length. Of that, 269 miles ran underground and 46 total miles aboveground; however, only about 36 miles consisted of arched structures — just under 12 percent in all.

الأسئلة حلها بنفسك أولًا، والإجابات تحتها

تفاصيل

51 Who developed rain harvesting in the third Millennium B.C.?

A Romans B Minoans C Greeks D Galicians

تفاصيل

52 How long were Rome's aqueducts in total?

A 46 miles B 100 miles C 269 miles D 315 miles

تفاصيل

53 When did Trajan build the Aqua Traiana?

A 312 B.C B 247 B.C C 109 A.D D 226 A.D

15 الإجابات مع الشرح

51 B Minoans

السبب المينويون في جزيرة كريت طوروا جمع مياه الأمطار في الألفية الثالثة قبل الميلاد.

the Minoans developed sophisticated rain-harvesting and filtering systems as early as the middle of the third millennium B.C.

52 D 315 miles

السبب الطول الكلي 315 ميلًا، منها 269 تحت الأرض و 46 فوقها.

Rome, for example, was supplied by aqueducts totaling 315 miles in length.

ستيب STEP: تجميعات متكررة

53 C 109 A.D

السبب من الجدول الزمني: سنة 109 ميلادية.

A.D. 109 Trajan builds the Aqua Traiana

ستيب STEP: تجميعات متكررة

16 سنغافورة
Singapore

ذكرها المختبرون 63 مرة
3 سؤال

(1) Though physically small, Singapore is an economic giant. It has been Southeast Asia's most modern city for over a century. The city blends Malay, Chinese, Arab, Indian and English cultures and religions. Its unique ethnic tapestry affords visitors a wide array of sightseeing and culinary opportunities from which to choose. A full calendar of traditional festivals and holidays celebrated throughout the year adds to its cultural appeal. In addition, Singapore offers luxury hotels, delectable cuisine and great shopping! The island nation of the Republic of Singapore lies one degree north of the Equator in Southern Asia. The country includes the island of Singapore and 58 or so smaller islands. Because of its efficient and determined government, Singapore has become a flourishing country that excels in trade and tourism and is a model to developing nations. The capital city, also called Singapore, covers about a third of the area of the main island.

(2) Located at the tip of the Malay Peninsula, Singapore's tropical climate welcomes both leisure and business travelers year-round. The island republic's excellent infrastructure enables visitors to enjoy its many sites and attractions in a safe, clean and green environment. Award winning Changi Airport provides airlinks to major cities around the world. The train and subway systems are clean, fast and efficient. In addition, its state-of-the-art cruise terminal has established Singapore as one of the premier cruising centers of South East Asia and an exciting port of call on any Asian cruise itinerary.

(3) In the city, there is no need for a car. Public transportation is excellent and walking is a good way to explore the city. All major attractions are also accessible by tour bus. Since the city is only 60 miles (100k) from the equator, the tropical temperatures do not vary much. Rainfall is fairly evenly distributed through the year. No matter when you choose to visit, warm weather will be abundantly available. The visitor is struck immediately by Singapore's abundance of parks, nature reserves, and lush, tropical greenery.

(4) Singapore's progress over the past three decades has been remarkable, yet the island has not been overwhelmed by development. Visitors will discover a wealth of historical treasures from the past, in the beauty of older buildings, values and traditions that have survived in the face of profound social and geographical change.

(5) Lacking any noteworthy natural resources, Singapore's early prosperity was based on a vigorous free trade policy, put in place in 1819 when Sir Stamford Raffles first established it as a British trading post. Later, mass industrialization bolstered the economy, and today the state boasts the world's second busiest port after Rotterdam, minimal unemployment, and a super-efficient infrastructure. Almost the entire population lives in upscale new apartments, and the average per capita income is over US$12,000. Singapore is a clean, safe place to visit, its amenities are second to none and its public places are smoke-free and hygienic.

(6) Forming the core of downtown Singapore is the Colonial District. Each surrounding enclave has its own distinct flavor, from the aromatic spice stores of Little India, to the tumbledown backstreets of Chinatown, where it is still possible to find calligraphers and fortune tellers, or the Arab Quarter,

whose cluttered stores sell fine cloths and silks.

(7) North of the city, are two nature preserves, Bukit Timah and the Central Catchment Area, along with the splendid Singapore Zoological Gardens. The east coast features good seafood restaurants set on long stretches of sandy beach. In addition, there are over fifty islands and islets within Singaporean waters, all of which can be reached with varying degrees of ease. Day trips are popular to Sentosa, the island amusement arcade which is linked to the south coast by a short causeway and cable car. Music, theater, nightlife: all are abundant in this remarkable city. Singapore used to be considered a "stop over" on the way to larger Asian cities. This is no longer true! Visitors seek out Singapore for business and finance and also for a fascinating and satisfying vacation for the whole family.

(8) Singapore is both an island and a country, but perhaps its best description is that of city-state. Like the great city-states of the past, it offers civilization and order in the highest degree. Its combination of Western-style development and Eastern style calm seems to present the best of both hemispheres: It's a modern metropolis where you feel safe walking the streets, and it's an Asian business center that's a model of efficiency. Singapore is also a multicultural city, and close to one-quarter of its population are expatriates or foreign workers from all over the world. Known for its desire to become the technology hub of Asia, Singapore is the most wired country in the region.

(9) Singapore shares another trait with historical city-states: Its authorities strongly believe that they can safeguard the status quo with regulations against almost anything and everything that-in their view-could possibly upset the sense of tranquility. In reality, visitors will find the place is not as restrictive as the long lists of hefty fines for such things as littering and jaywalking suggest. Some visitors to Singapore leave singing the praises of a society that "works," while others feel the government's near-compulsive fixation on cleanliness and order makes Singapore sterile in every sense of the word.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

54 **In what two areas is Singapore a successful country?** تفاصيل

- **A** management of natural resources and trade
- **B** architecture and public transport
- **C** business and finance
- **D** trade and tourism

55 **What two attractions are found north of the city of Singapore?** تفاصيل

- **A** Singapore Zoological Gardens and the Arab Quarter
- **B** Bukit Timah and the Central Catchment Area
- **C** Timah Sentosa and the Colonial District
- **D** Little India and Bukit

تفاصيل

56 **In the early days, what helped strengthen Singapore's economy?**

- **A** a strong trade policy and industrial development
- **B** low unemployment and a free trade policy
- **C** natural resources and a busy trading port
- **D** tourism and industrial development

16 الإجابات مع الشرح

54 **D trade and tourism**

السبب الفقرة 1 تقول صراحة إنها تتفوق في التجارة والسياحة.

> *Singapore has become a flourishing country that excels in trade and tourism*

55 **B Bukit Timah and the Central Catchment Area**

السبب الفقرة 7: شمال المدينة محميتان طبيعيتان.

> *North of the city, are two nature preserves, Bukit Timah and the Central Catchment Area*

56 **A a strong trade policy and industrial development**

السبب في البداية: سياسة تجارة حرة قوية، ثم دعمها التصنيع.

> *Singapore's early prosperity was based on a vigorous free trade policy ... Later, mass industrialization bolstered the economy*

ليه الباقي غلط C غلط لأن سنغافورة بلا موارد طبيعية تُذكر، والبطالة المنخفضة اليوم وليست في البداية.

ستيب STEP تجميعات متكررة

17

مصادر الطاقة المتجددة
Renewable Energy Sources (Passages A and B)

ذكرها المختبرون 62 مرة

2 سؤال

Passage A

Solar Energy

(1) The energy falling on the Earth from the Sun is mostly in the form of light and the light that falls in one hour equals the total energy used by the world in a year, unfortunately, its low energy density requires large collecting devices and its availability varies. Its greatest potential use is as an energy source for low-temperature water heating. This uses solar panels as the energy transfer devices, which convert light into heat energy. They are used increasingly to produce domestic hot water at about 70°C and to heat swimming pools

(2) Solar energy can also be used to produce high temperature heating, up to 3000°C or so, if a large curved mirror (a solar furnace) focuses the Sun's rays on to a small area. The energy can then be used to turn water to steam for driving the turbine of an electric generator in a power station

(3) Solar cells, made from semiconducting materials, convert sunlight into electricity directly. A number of cells connected together can be used to supply electricity to homes and to the electronic equipment in communication and other satellites. They are also used for small-scale power generation in remote areas of developing countries where there is no electricity supply. Recent developments have made large-scale generation more cost effective and there is now a large solar power plant in California. There are many designs for prototype light vehicles run on solar power.

Wind energy

(4) Giant windmills called wind turbines with two or three blades each up to 30m long drive electrical generators. "Wind farms" of 20 to 100 turbines spaced about 400m apart, supply about 400MW (enough electricity for 250,000 homes) in the UK and provide a useful "top up" to the National Grid. Wind turbines can be noisy and may be considered unsightly so there is some environmental objection to wind farms, especially as the best sites are often in coastal or upland areas of great natural beauty.

Passage B

Wave energy

(5) The rise and fall of sea waves has to be transferred by some kind of wave-energy converter into the rotary motion required to drive a generator. It is a difficult problem and the large-scale production of electricity by this means is unlikely in the near future, but small systems are being developed to supply island communities with power.

Tidal and hydroelectric energy

(6) The flow of water from a higher to a lower level from behind a tidal barrage (barrier) or the dam of a hydroelectric scheme is used to drive a water turbine (water wheel) connected to a generator.

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(7) One of the largest working tidal schemes is the La Rance project in Canada. Feasibility studies have shown that a 10 - mile-long barrage across the River Severn Estuary could produce about 7% of today's electrical energy consumption in England and Wales. Such schemes have significant implications for the environment, as they may destroy wildlife habitats of water birds for example, and also for shipping routes.

(8) In the UK, hydroelectric power stations generate about 2% of the electricity supply. Most are located in Scotland and Wales where the average rainfall is higher than in other areas. With good management hydroelectric energy is a reliable energy source, but there are risks connected with the construction of dams, and a variety of problems may result from the impact of a dam on the environment. Land previously used for forestry or farming may have to be flooded.

الأسئلة حلها بنفسك أولًا، والإجابات تحتها

57 الفكرة الرئيسية

Which of the following statements best summarizes renewable energy from both passages?

A The UK is the leading country in the world using alternative energy sources
B Renewable energy make up a small portion of the energy supply
C wind turbines cause too much environmental damage to be used widely
D Canada is the leading country is hydroelectric energy

58 تفاصيل

What do the passages say about the UK?

A Wind turbines are unpopular in the UK, but they provide 400 MW of energy
B the UK has up to 100 wind turbines that can provide energy to other countries
C Wind turbines supply electricity to some homes and hydroelectricity powers 2 percent of homes in the UK.
D Hydroelectricity powers 2 percent of homes and wind power could provide power for another

17 الإجابات مع الشرح

57 **B Renewable energy make up a small portion of the energy supply**

السبب الطاقة المتجددة في القطعتين «مكمّل» صغير: الرياح «إضافة مفيدة» للشبكة، والطاقة المائية 2% فقط.

provide a useful "top up" to the National Grid ... hydroelectric power stations generate about 2% of the electricity supply

58 **A** **Wind turbines are unpopular in the UK, but they provide 400 MW of energy**

السبب هناك اعتراض على مزارع الرياح، ومع ذلك تعطي 400MW.

> *supply about 400MW (enough electricity for 250,000 homes) in the UK ... so there is some environmental objection to wind farms*

ليه الباقي غلط C و D غلط: الطاقة المائية تنتج 2% من الكهرباء وليس من المنازل.

ستيب STEP: تجميعات متكررة

18

فراولة كاليفورنيا
California Strawberries

ذكرها المختبرون 59 مرة

2 سؤال

(1) Agricultural abundance is a pillar of the California dream. In 2016 the state turned out more than $45 billion worth of meat, milk and crops. Long before nutritionists agreed that fresh fruits and vegetables should be the center of American diets, California farmers had planted much of their land in these products, and today they produce half of the nation's fruits, vegetables and nuts.

(2) But although fruits and vegetables are vaunted as healthy foods, their impact as crops is quite different. On many California produce farms wages are low, working conditions are poor and farmers use enormous quantities of pesticides and precious water. This is the central contradiction of California agriculture.

(3) California's strawberry industry is the state's sixth most important commodity in terms of the value of crops sold. Strawberries are attractive reasonably nutritious and occasionally tasty fruits and can be grown and eaten within California nearly year-round. But the industry's growth has relied on heavy use of toxic chemicals and now growers face heightened restrictions on some of their most favored chemicals; soil fumigants.

(4) Unfortunately, less toxic or non-chemical strategies that would allow strawberries to be grown for a mass market, maintaining affordable prices, are elusive and likely to remain so.

(5) Although strawberry production once was scattered throughout the state, by the 1960s, it had concentrated in coastal zones to take advantage of sandy soils and mild temperatures. Thereafter, the industry saw tremendous growth in productivity. In Monterey and Santa Cruz counties alone, acreage more than tripled and production increased tenfold from 1960 to 2014. Much of this growth was enabled by advances in plant breeding and use of plastic tarps to absorb heat, allowing growers to increase the length of strawberry seasons.

(6) But the main driver of growth has been the use of pre plant chemical fumigants. Growers hire pest control companies to fumigate soils before planting strawberries in order to kill soil-borne pests most importantly, plant pathogens such as Verticillium dahliae and Macrophomina phaseolina Without such treatment, these pathogens cause strawberry plants to wilt and die.

(7) Now, however, the industry's fumigant of choice methyl bromide can no longer be used in strawberry fruit production. In 1991 methyl bromide was banned under the Montreal Protocol on Substances that used in nursery production to ensure that starter plants are virus-and pathogen-free.

(8) One potential replacement, methyl iodide, was approved for use in late 2010. But it was withdrawn from the market in 2012, following an activist campaign and lawsuit that accused California regulators of performing an inadequate review of potential health risks to workers and the general public. Among other things, the chemical is a known neurotoxin and carcinogen.

(9) Other fumigants are still allowed, but their use is increasingly restricted by buffer zones and township quotas. Consequently, growers are contending with heightened levels of plant disease, some from pathogens that had never before been evident in California strawberry fields.

ملف مجاني من تطبيق ستيب — Google Play و App Store

(10) Can California find a less toxic way to raise 90 percent of the nation's fresh strawberries? Although the strawberry industry is investing significant resources into non-chemical alternatives to manage soil-borne disease, the obstacles are formidable. The entire production system, including reliance on fumigants, is embedded into the cost of land.

(11) Fumigation has allowed growers to plant on the same blocks of land. year after year, and not worry about soil disease. With fumigation available to control pathogens. Strawberry breeders have emphasized productivity, beauty and durability. Meanwhile, nursery production has shifted away from prime fruit growing regions along the coast to take advantage of different environments for plant propagation, enabling coastal land to be used solely for growing fruit.

(12) Together these innovations have allowed growers to keep prime strawberry land in production every year for much of the year, yielding exceptional amounts of fruit. High land prices reflect these expectations and make it unprofitable to grow strawberries using less intensive methods. The Pacific Ocean's natural summer air-conditioning is attractive to suburbanites as well as strawberries, so coastal development is putting additional pressure on the cost of strawberry land while at the same time increasing public pressure to control use of fumigants.

(13) Informed and concerned consumers ingrained with California's deep culture of environmentalism have turned to organic strawberries, which they see as a more sustainable option. As conventional growers took note of this vibrant market, organic strawberry production rose fivefold between 2000 and 2012, to reach about 3,300 acres planted in 2017, which represents 12 percent of all strawberry acreage.

(14) But although organic growers use non-chemical soil fumigation methods or rotate strawberries with crops that have a mild disease-suppressing effect, such as broccoli, few of them fundamentally alter the production system in other ways.

(15) A small but dedicated set of growers have learned how to raise strawberries for the long haul without fumigants. However, even they use starter plants produced on fumigated soil, since no nurseries produce organic plants. Crucially, for these growers strawberries are a minor crop in what are otherwise highly diversified systems. And most of these producers are located outside of prime strawberry growing regions, where land is cheaper. Their approach therefore is not nearly replicable for growers producing for the mass market.

(16) These exceptions tell us as much about the limits of California strawberry production as does mainstream production. Consumers who want organic strawberries must be willing to live with compromises, pay premium prices. For others, the dream of affordable year-round strawberries grown without toxic chemicals is already an impossible one.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

59 **Between which years did organic strawberry production rise by fivefold?** تفاصيل

A 1960 and 2014
B 2005 and 2015
C 2010 and 2012
D 2000 and 2012

ملف مجاني من تطبيق ستيب — Google Play و App Store

تفاصيل

60 **What are two plant pathogens that harm strawberries?**

- **A** verticillium dahliae and macrophomina phaseolina
- **B** methyl bromide and methyl iodide
- **C** methyl iodide and macrophomina phaseolina
- **D** verticillium dahliae and methyl bromide

18 الإجابات مع الشرح

59 **D 2000 and 2012**

السبب الفقرة 13: بين 2000 و 2012.

> *organic strawberry production rose fivefold between 2000 and 2012*

60 **A verticillium dahliae and macrophomina phaseolina**

السبب الفقرة 6 تسمي المُمرضين، أما methyl bromide و methyl iodide فهما مواد تبخير.

> *plant pathogens such as Verticillium dahliae and Macrophomina phaseolina*

ستيب STEP: تجميعات متكررة

19 الإعلانات Advertising

ذكرها المختبرون 58 مرة
3 سؤال

(1) Whenever people give information to the public about an event, a product, or a service, they are using advertising. A piece of advertising is called an advertisement, or ad. Ads appear almost everywhere. They are found in newspapers and magazines. They appear on television, radio, and the Internet. Ads are also posted as signs in all kinds of public places.

(2) The goal of most advertising is to make the public want to buy whatever is being advertised. People also use advertising to make the public form opinions about things. For example, an ad could try to make the public believe that a company treats its workers well.

(3) When advertisers want many people to see or hear their ads, they pay the media to have the ads published or broadcast. Newspapers and magazines publish, or print ads between their articles. Television and radio network broadcast, or play ads during their programs. Television and radio ads are also called commercials. Web sites, buses, movie theaters, and many other places also display ads in exchange for money.

(4) Signs were the fast form of advertising. Even thousands of years ago people created hand-lettered signs to advertise their goods and services. The invention of the printing press in the 1400s expanded this type of advertising. Merchants could print dozens of the same posters and fliers to advertise what they were selling. In the 1600s and 1700s newspapers appeared in Europe. The Americas and Japan. Advertising then became even more widespread.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

61 تفاصيل

What is one important idea related to advertisements that the writer mentions in Paragraph 2?

A that they are everywhere
B that they affect our views
C that they give information
D that they help businesses

62 تفاصيل

What is one important idea related to advertisements that the writer mentions in paragraph (1)?

A that they existed thousands of years ago
B that they make us want to buy things
C that they may change our views
D that they are found in all places

ملف مجاني من تطبيق ستيب — Google Play و App Store

استنتاج

63 **What does the writer want to show when he mentions that an ad could make us believe that company treats its workers well?**

A that ads can benefit businesses
B that ads can convince us to buy
C that ads can shape our opinions
D that ads can give us information

19 الإجابات مع الشرح

61 **B that they affect our views**

السبب: الفقرة 2: الإعلان يجعل الناس يكوّنون آراء عن الأشياء.

People also use advertising to make the public form opinions about things.

62 **D that they are found in all places**

السبب: الفقرة 1: الإعلانات تظهر في كل مكان تقريبًا.

Ads appear almost everywhere.

ليه الباقي غلط: B و C من الفقرة 2، و A من الفقرة 4.

63 **C that ads can shape our opinions**

السبب: المثال جاء بعد جملة تكوين الآراء مباشرة، فهو يوضح أن الإعلان يشكّل آراءنا.

For example, an ad could try to make the public believe that a company treats its workers well.

ستيب STEP: تجميعات متكررة

20 الدمية الروسية
Matryoshka (Russian Doll)

ذكرها المختبرون 58 مرة
2 سؤال

(1) A matryoshka doll refers to a set of dolls of decreasing size placed one inside the other. The word “matryoshka”, literally “little matron”, is a diminutive form of the Russian female first name “Matryona”. The first doll set in Russia was carved in 1890 by Vasily Zvyozdochkin from a design by Sergey Malyutin, who was a folk crafts painter. Traditionally the outer layer is a woman, dressed in a sarafan, a long traditional Russian dress.

(2) It is believed Zvyozdochkin and Malyutin were inspired by a doll from Honshu, the main island of Japan. Sources differ in their descriptions, describing it as either a round, hollow doll or a nesting doll, portraying a bald old Buddhist monk.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

64 The best title for this passage is _____ — الفكرة الرئيسية

A Arts and Crafts
B Children's Toys
C Modern Handicrafts
D Russian Wood figurines

65 According to Paragraph (2), the idea of nesting dolls most likely started in _____ — استنتاج

A Russia
B Japan
C Paris
D America

20 الإجابات مع الشرح

64 D Russian Wood figurines

السبب: القطعة كلها عن الدمية الروسية المنحوتة carved من الخشب.

The first doll set in Russia was carved in 1890 by Vasily Zvyozdochkin

65 B Japan

السبب: صانعا الدمية استوحياها من دمية من جزيرة هونشو في اليابان.

It is believed Zvyozdochkin and Malyutin were inspired by a doll from Honshu, the main island of Japan.

ستيب STEP تجميعات متكررة

ملف مجاني من تطبيق ستيب — Google Play و App Store

21

الإعلانات بين المؤيدين والمنتقدين
Advertising: Defenses and Criticisms

ذكرها المختبرون 58 مرة

3 سؤال

Advertising

Defenses of Advertising

(1) Defenders of advertising claim that it provides many benefits. It informs consumers about available products and services. Some adverts promote the public interest by informing consumers about health and safety issues. Volvos, for example, were among the first vehicles to provide side-door air bags. This is a definite safety advantage. Defenders of advertising also argue that by giving media the financial backing it requires, advertisements allow "free" television and internet content. The survival and profitability of many media outlets, such as social media, television and radio stations and YouTube channels depend on advertising revenue. Finally, advocates of advertising claim that it stimulates competition and fuels our mass consumption economy. It raises the standard of living for everyone because it makes economies of large-scale production possible. Mass production reduces the cost of manufacturing a product, so it is less expensive to buy; but mass production requires mass consumption, and mass consumption of a product cannot occur unless consumers know that the product exists and where it can be bought. So, advertising is required if we are to take advantage of the benefits of large-scale production

Criticisms of Advertising

(2) Numerous criticisms have been levelled at advertising. First, some critics argue, advertising interferes with almost everything we do (studying, reading, driving, watching YouTube, browsing the internet). Second, critics say, advertising makes citizens materialistic. They only care about possessing more and more. Of course, there is nothing wrong with improving your standard of living, but they say advertising encourages consumers to measure their worth only by what they have, rather than by who they are. And so, advertising weakens social bonds and contributes to a shallow, materialistic culture. Third, advertising exploits children by targeting dubious messages (sugary cereals are good, cool kids wear brand-name clothing) at an audience that is too young to be able to evaluate their content. ...

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

66 معنى كلمة

What are two words that the writer uses for “supporter “ in paragraph (1)?

A consumer and defender
B consumer and survival
C advocate and survival
D advocate and defender

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67 **Which of the following is an opinion?** استنتاج

A Large scale production makes products cheaper
B Some TV stations need revenue from advertising
C Advertising makes people want more possessions
D Advertising gives people information about products.

68 **What can we understand about YouTube from the passage?** استنتاج

A People use it to make money
B It is popular with young people
C It stops students from studying
D It has more adverts than television

21 الإجابات مع الشرح

66 **D advocate and defender**

السبب الكاتب يسمي المؤيدين defenders و advocates.

Defenders of advertising claim ... advocates of advertising claim

67 **C Advertising makes people want more possessions**

السبب هذا قول المنتقدين (critics say)، وهو رأي يحتمل الخلاف؛ باقي الخيارات معلومات يمكن التأكد منها.

Second, critics say, advertising makes citizens materialistic.

68 **A People use it to make money**

السبب قنوات يوتيوب تعتمد في ربحها على دخل الإعلانات، أي أن الناس يكسبون منها المال.

The survival and profitability of many media outlets, such as social media, television and radio stations and YouTube channels depend on advertising revenue.

ملف مجاني من تطبيق ستيب — Google Play و App Store

22 الساعات Watches

ذكرها المختبرون 56 مرة

3 سؤال

A watch is a portable timepiece that is designed to be worn on the wrist or carried in the pocket. The movement of a watch is driven either by a spring or by electricity. Along with clocks, watches are the most common devices for measuring time. Watches were developed later than clocks. The first portable timekeeper or watch was produced during the Renaissance. A weight and a crude escapement were suitable in clocks but could not be used in a portable watch. In the early 1500s a coiled metal spring that transmitted energy as it unwound was substituted for the weight. By the 1580s, about the time of Galileo's pendulum experiments, the first commercial watches were being produced in small quantities in Nuremberg (Nunberg), Germany. Known as Nuremberg eggs, these early watches were portable but had only an hour hand because the escapement mechanism was too inaccurate to show minutes.

The English physicist Robert Hooke, who introduced the basic pendulum and escapement mechanisms in clocks in the 1660s, applied the same idea to watches. He attached a small balance unit to the mainspring unit. A balance spring, or hairspring, was used to control the oscillations of the balance unit. This design formed the basis for all subsequent mechanical watches.

The mechanism of mechanical watch is made up of six different parts. These are the mainspring, which is the source of energy; the gear train, which transmits the energy; the dial train, which governs the movement of the hands; the winding and setting mechanism; the escapement and balance unit, which controls the release of energy; and the plates, or framework, which enclose and protect the movement.

In operation the mainspring unwinds to turn the barrel, which drives the escape, or great, wheel through a gear train. The wheel moves in one direction until stopped by a pallet arm of the pallet and fork – the part analogous to the detent of a clock. The pallet and fork assembly is attached to the balance wheel, which is then given an impulse from the escape wheel.

The escapement and balance assembly are essential to an accurate watch mechanism because they control the release of energy from the mainspring and the time in which each release of energy is made. The oscillations of the balance, regularity of its movements, are controlled with a hairspring or escapement spring. When the balance wheel moves in one direction, its tiny hairspring is wound. As the escape wheel is stopped by one of the pallet arms, the hairspring uncoils. The balance wheel, in turn, is connected to the coiled mainspring, which winds as the balance wheel moves. When the pallet and fork assembly releases the escape wheel, the wheel rotates one tooth until caught again by the other pallet arm. The balance wheel then turns in the opposite direction until the other pallet arm stops the escape wheel. This cycle continues and causes the familiar ticking sound of the mechanical watch. The escape wheel is stopped and released every one-fifth of a second. Thus, the balance wheel swings back and forth 300 times a minute, or 432,000 times a day. As in a clock, a series of gears transfers the motion of the escape wheel to the hour, minute, and second hands.

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Electric-powered watches use one of three drive systems: (1) the galvanometer drive, consisting of the conventional balance-hairspring oscillator, kept in motion by the magnetic interaction of a coil and a permanent magnet, (2) the induction drive, in which an electromagnet attracts a balance containing soft magnetic material, or (3) the resonance drive, in which a tiny tuning fork (about 1 inch [25 millimeters] in length), driven electrically, provides the motive power. Both galvanometer and induction drive types use a mechanical contact, actuated by the balance motion, to provide properly timed electric-drive pulses. Each oscillation of the balance operates a time-indicating gear train by advancing a toothed wheel one tooth.

الأسئلة حلها بنفسك أولًا، والإجابات تحتها

تفاصيل

69 What starts the movement of a watch?

A A spring or a pendulum
B Electricity or a weight
C A spring or a weight
D Electricity or a spring

تفاصيل

70 What controls the release of energy in a mechanical watch?

A The gear and dial train
B The plates and balance unit
C The winding and setting system
D The balance and escapement unit.

تفاصيل

71 Which two types provide timed electric-drive pulses?

A The galvanometer and induction drive
B The induction and resonance drive
C The electronic and induction drive
D The galvanometer and resonance drive

22 الإجابات مع الشرح

69 D Electricity or a spring

السبب الجملة الثانية في القطعة: نابض أو كهرباء.

The movement of a watch is driven either by a spring or by electricity.

ليه الباقي غلط الثقل weight كان في الساعات الكبيرة ولا يصلح في ساعة محمولة.

70 D The balance and escapement unit.

السبب الميزان ونظام الإفلات هما اللذان يتحكمان في إطلاق الطاقة.

the escapement and balance unit, which controls the release of energy

ملف مجاني من تطبيق ستيب — Google Play و App Store

71 **A** The galvanometer and induction drive

السبب النوعان galvanometer و induction يستخدمان تلامسًا ميكانيكيًا لنبضات الكهرباء.

Both galvanometer and induction drive types use a mechanical contact, actuated by the balance motion, to provide properly timed electric-drive pulses.

ستيب STEP: تجميعات متكررة

23 الإنفلونزا الإسبانية
The Spanish Flu

ذكرها المختبرون 48 مرة
5 سؤال

From 1918 to 1920 the virus known as the Spanish Flu infected 500 million people worldwide, killing 50 to 100 million of them. The only continent it did not reach was Antarctica; deaths were registered from northern Alaska to the islands of Samoa. The virus began in conditions that were perfect to help it spread across the globe, turning the battlefields of World War I into huge centers of infection. Though the flu had long been a familiar disease, the subtype (special type) of 1918 took scientists by surprise. They were shocked as much by the speed of its spread as by the number of people it killed. Though some experts believe the virus's origins were in Europe, the theory most accepted today suggests that it began in the United States, and later travelled on an army ship to the battlefields of France. From there, an army of disease germs, like enemy soldiers in a secret war 'Ashgan', spread out through the air and infected a third of the world's population, killing between 2.5% and 5% of everyone on earth. In the future, suggests the World Health Organization, there will be more and perhaps more deadly flu pandemics (worldwide diseases), threatening global health and security and putting the future of human society at risk. The virus won't need a great war to help it cover the earth; it will be enough for one carrier to get on an airplane and, in less than 20 hours, cover the distance between Tokyo and New York, or between Sydney and Doha.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

72 معنى كلمة

What phrase other than "Spanish Flu" does the writer use to describe the deadly virus of 1918?

A enemy soldiers
B World War I
C familiar disease
D centers of infection

73 استنتاج

Which of the following statements about the Spanish Flu is an opinion?

A It began in the United States.
B It killed at least 50 million people.
C There were no infections in Antarctica.
D Now it could spread across the world in one day.

ستيب STEP: تجميعات متكررة

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استنتاج

74 **How is the view of some experts about the origins of the Spanish Flu different from the most common view?**

A They believe there were two types of the virus.
B They believe it started in the United States.
C They believe it was a biological weapon.
D They believe it began in Europe.

استنتاج

75 **What can we understand from the passage about World War I?**

A It was a secret war.
B It was the origin of the Spanish Flu.
C It increased deaths from the Spanish Flu.
D It killed between 50 and 100 million people.

استنتاج

76 **What can we understand from the passage about the Spanish Flu?**

A Governments lied about it.
B It would spread faster now.
C It will come back in the future.
D Fewer people would die from it now.

23 الإجابات مع الشرح

72 **A enemy soldiers**

السبب الكاتب يشبّه الجراثيم بجنود أعداء في حرب سرية.

an army of disease germs, like enemy soldiers in a secret war

73 **D Now it could spread across the world in one day.**

السبب هذا توقع للمستقبل (suggests) وليس حدثًا وقع، فهو رأي؛ باقي الخيارات معلومات تاريخية.

it will be enough for one carrier to get on an airplane and, in less than 20 hours, cover the distance between Tokyo and New York

74 **D They believe it began in Europe.**

السبب بعض الخبراء يرون أنه بدأ في أوروبا، والرأي الأشهر أنه بدأ في أمريكا.

Though some experts believe the virus's origins were in Europe, the theory most accepted today suggests that it began in the United States

ليه الباقي غلط B هو الرأي الشائع نفسه، وليس رأي «بعض الخبراء».

ملف مجاني من تطبيق ستيب — App Store و Google Play

75 **C** It increased deaths from the Spanish Flu.

السبب ساحات المعارك صارت مراكز عدوى ضخمة، أي أن الحرب زادت الإصابات والوفيات.

> *turning the battlefields of World War I into huge centers of infection*

76 **B** It would spread faster now.

السبب اليوم يكفي أن يركب شخص مصاب طائرة ليصل المرض لقارة ثانية في أقل من 20 ساعة.

> *The virus won't need a great war to help it cover the earth*

ليه الباقي غلط C غير دقيق: القطعة تتوقع أوبئة جديدة، لا عودة الإنفلونزا الإسبانية نفسها.

ستيب STEP: تجميعات متكررة

24 الأتمتة Automation

ذكرها المختبرون 48 مرة

6 سؤال

Automation

(1) Knowing how and where robots and machines could impact the global economy will help financial executives gain a competitive edge in their industries. But predicting which countries are likely to be impacted most by workforce automation and when isn't as simple as comparing industries across the globe.

(2) A recent McKinsey Global Institute report, A Future That Works: Automation, Employment, and Productivity, analyzed the automation potential of 46 countries, representing 80% of the global workforce. Several factors are considered, including the percentage of work activities that could be automated using current technology, the number of full-time employees that could be affected and wages.

(3) The types of activities that have a high potential for automation are physical tasks in highly structured and predictable environments, data processing and data collection. Those that have a considerably lower potential for automation are unpredictable physical work, interactions with others, applying expertise, and managing others, which is the least susceptible to automation.

(4) Vast differences are expected in how automation will play out across the globe, as technical, economic, and social factors will determine the pace and extent of it. But it's hard to say how quickly automation will become reality, according to the report.

(5) Take manufacturing, for example. One country may have a larger concentration of work hours in jobs that have higher automation potential, such as production and administrative support. And another country may have a higher proportion of work hours in jobs that are less likely to be automated, such as management and engineering.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

77 معنى كلمة

What idea do the words "susceptible to automation" repeat?

A workforce automation
B robots and machines
C automation potential
D competitive edge

78 معنى كلمة

What idea do the words "administrative support" repeat?

A Management and engineering
B Data processing and data collection
C Technical, economic and social factors.
D Applying expertise and managing others

ملف مجاني من تطبيق ستيب — Google Play و App Store

استنتاج

79 **What can we understand from Paragraph (2) about the McKinsey Global Institute research?**

A It was a large study
B It was an automated study
C It used current technology
D It used an effect on full-time employees

استنتاج

80 **Which of the following is an opinion in the passage?**

A The report considered several factors in the automation process
B Automation of work will become a reality for all workers.
C The speed of change to automation is difficult to predict.
D The McKinsey report studied 46 different countries

استنتاج

81 **Which of the following is a fact?**

A physical work can be unpredictable
B The report considered information from 46 countries
C Technology has the greatest effect on full-time employees
D The writers of the report interviewed 80% of the global workforce

استنتاج

82 **What can we understand from Paragraphs (4) and (5) about automation around the world?**

A It will be very difficult to achieve
B It will become a reality very quickly
C It will be different in different countries
D It will have the biggest effect on manufacturing.

24 الإجابات مع الشرح

77 **C automation potential**

السبب susceptible to automation = قابل للأتمتة، وهي نفس فكرة automation potential.

Those that have a considerably lower potential for automation are unpredictable physical work, interactions with others, applying expertise, and managing others, which is the least susceptible to automation.

ملف مجاني من تطبيق ستيب — Google Play و App Store

78 **B** Data processing and data collection

السبب الدعم الإداري من الوظائف عالية القابلية للأتمتة، مثل معالجة البيانات وجمعها.

jobs that have higher automation potential, such as production and administrative support

ليه الباقي غلط A و D وظائف قليلة القابلية للأتمتة، عكس المطلوب.

79 **A** It was a large study

السبب الدراسة شملت 46 دولة تمثل 80% من القوى العاملة في العالم، فهي دراسة كبيرة.

analyzed the automation potential of 46 countries, representing 80% of the global workforce

80 **C** The speed of change to automation is difficult to predict.

السبب «من الصعب معرفة السرعة» تقدير وليس معلومة ثابتة، فهو رأي.

But it's hard to say how quickly automation will become reality, according to the report.

81 **B** The report considered information from 46 countries

السبب عدد الدول (46) معلومة يمكن التأكد منها = حقيقة.

analyzed the automation potential of 46 countries

ليه الباقي غلط D غلط: الـ 80% نسبة القوى العاملة التي تمثلها الدول، ولم يُقابَلوا.

82 **C** It will be different in different countries

السبب الأتمتة ستختلف من دولة لأخرى حسب نوع الوظائف فيها.

Vast differences are expected in how automation will play out across the globe

ستيب STEP تجميعات متكررة

25

حديقة يلوستون الوطنية
Yellowstone National Park

ذكرها المختبرون 47 مرة

3 سؤال

(1) Established in 1872, Yellowstone National Park is not only the oldest national park in the United States but also in the world. It is also one of the largest, and probably the best-known, national parks in the United States. It is located principally in northwestern Wyoming and partly in southern Montana and eastern Idaho. The park, which forms a square like rectangle, is 63 miles (101 kilometers) from north to south and 54 miles (87 kilometers) from east to west at its widest point; it covers an area of 3,472 square miles (8.992 square kilometers). Grand Teton National Park lies to the south, and on all other sides Yellowstone is surrounded by national forests. Yellowstone's headquarters are at Mammoth Hot Springs near the northern entrance to the park.

(2) Most of the park consists of broad volcanic plateaus with an average elevation of about 7,875 feet (2,400 meters). Three mountain ranges, each running roughly north to south, protrude into the park: the Gallatin Range in the northwest, the Absaroka Range in the east, and the northern extremity of the Teton Range along the park's southwestern boundary. The tallest mountains in the park are in the Absarokas, where many summits exceed elevations of 10.000 feet (3,050 meters). The range's Eagle Peak, on the park's boundary in the southeast is the high point reaching 11,358 feet (3.462 meters). Within its rugged mountains are spectacular deep glacier-carved valleys.

(3) Yellowstone is also known for its many scenic lakes and rivers. The park's largest body of water is Yellowstone Lake, which has a surface area of 132 square miles (342 square kilometers) and lies at an elevation of 7,730 feet (2.356 meters), making it the highest mountain lake of its size in North America. The Yellowstone River enters at the southeast corner, flows generally northward (including through Yellowstone Lake). and exits near the northwest corner of the park. The river's Yellowstone Falls, located in the north-central part of the park, descend in two majestic cascades. The falls constitute the western end of the spectacular Grand Canyon of the Yellowstone. There the river has cut a gorge 19 miles (30 kilometers) long, between 800 and 1,200 feet (240 and 370 meters) deep, and up to 4.000 feet (1.200 meters) wide. The walls of the canyon are brilliantly colored in hues of red, pink, yellow, buff, lavender, and white.

(4) Yellowstone's other noted geysers and thermal features include Giantess Geyser, with a two-to six-month wait between eruptions: Excelsior Geyser, which rarely erupts but discharges thousands of gallons of boiling water per minute; and the Fountain Paint Pots, with pink, plopping mud geysers, fumaroles, and a blue hot-spring pool. Steamboat Geyser can throw water to heights of 300 feet (90 meters) and higher and is the world's highest-erupting geyser. Mammoth Hot Springs consists of abroad terraced hill side of travertine (calcium carbonate) deposited there by dozens of hot springs: some of the dazzling white rock is tinted by microorganisms on the rock.

(5) Yellowstone's climate is influenced by the park's high average elevation, location deep within the continent, and mountainous terrain. Temperatures almost always decrease with elevation gain. Because of Yellowstone's large area and variegated topography, weather conditions at different locations within the park can vary widely at any given time. In addition, conditions can change dramatically in a short period of time at one location.

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(6) Summer days are warm and relatively sunny, with daytime temperatures reaching about 80° F (27° C) in July at lower elevations before dropping to nighttime lows in the 40s or 50s F (about 10° C); temperatures are usually cooler higher in the mountains. Although precipitation totals in the warm months generally are light, summer afternoon thunderstorms are common. Winters are cold and snowy, as temperatures rarely rise above the mid-20s F (about-4° C) and often drop to 0° F (-18° C) or lower at night. Snowfall is heavy in most areas, especially at higher elevations, and typically begins in early autumn and continues into April or May.

(7) About four-fifths of the park's area is forested, and the vast majority of the tree growth consists of lodgepole pines. Among the several other conifer species in the park are white bark pine, found at higher elevations; and Douglas firs, which dominate at lower elevations. Cottonwoods and willows grow along streams, and stands of aspens occur in many sections. A significant event in Yellowstone's more recent history was the great fire of 1988 that charred some two-fifths of the park's forestlands. The forests subsequently have recovered, although areas have been subject to damage by insect pests. Hundreds of types of wildflowers thrive in a variety of habitats. Common species include phlox, lupines, cinquefoils, larkspurs, and Indian paintbrushes.

(8) Yellowstone boasts the greatest assemblage of mammals-more than five dozen species-in the United States, outside of Alaska. Bison (buffalo), the largest of the mammals, were brought back from near extinction at the beginning of the 20th century. Other large mammals often seen in Yellowstone include elk (wapiti), mule deer, black bears, foxes, and coyotes. There are smaller populations of brown. (grizzly) bears, bighorn sheep, pronghorns, mountain goats, and moose. Bobcats are thought to be present throughout the park, but their numbers are unknown, and there are occasional sightings of lynx and pumas (mountain lions). Common smaller mammals include badgers, martens, weasels, river otters, hares and rabbits, shrews, a variety of bats, and many species of small rodents. Wolves were successfully reintroduced into Yellowstone in 1995 and are now found throughout much of the park. Beavers have also made a significant comeback, and several hundred live along streams and lakes.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

83 **What are two features of the Yellowstone National Park?**

تفاصيل

A oldest national park in the world and highest mountain lake in the USA.
B highest mountain lake in the USA and 54-mile (87 km) gorge.
C 45-mile (87 km) gorge and 3 dozen animal species.
D a pink hot spring pool and 4 dozen animal species.

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تفاصيل

84 **What damage occurred in Yellowstone National Park?**

A fire and insect pests.
B fire and water pollution.
C insect pests and large amount of trash.
D water pollution and large amount of trash.

تفاصيل

85 **What types of trees are found in Yellowstone National Park?**

A oak and lupine.
B Douglas fir and oak.
C cottonwood and redwood.
D cottonwood and Douglas fir.

25 الإجابات مع الشرح

83 **A oldest national park in the world and highest mountain lake in the USA.**

السبب: أقدم حديقة وطنية في العالم، وبحيرتها أعلى بحيرة جبلية بحجمها في أمريكا الشمالية.

> *not only the oldest national park in the United States but also in the world ... making it the highest mountain lake of its size in North America*

ليه الباقي غلط: طول الوادي 19 ميلًا (و 54 ميلًا عرض الحديقة)، والثدييات «أكثر من خمس دزينات».

84 **A fire and insect pests.**

السبب: حريق 1988 أحرق خُمسي الغابات، ثم تضررت مناطق من الآفات الحشرية.

> *the great fire of 1988 that charred some two-fifths of the park's forestlands ... areas have been subject to damage by insect pests*

85 **D cottonwood and Douglas fir.**

السبب: الفقرة 7 تذكر Douglas firs و Cottonwoods.

> *Douglas firs, which dominate at lower elevations. Cottonwoods and willows grow along streams*

ليه الباقي غلط: oak و redwood غير مذكورة، و lupine زهرة وليست شجرة.

ستيب STEP تجميعات متكررة

26

التذوق والشم
Taste and Smell (Passages A and B)

ذكرها المختبرون 46 مرة

3 سؤال

Passage A: Taste

(1) Our sense of taste involves four basic sensations-sweet, sour, salty, and bitter. All other tastes are mixtures of these. Investigators have, however, been frustrated in their search for specialized nerve fibers for each of the four basic taste sensations.

(2) Taste is a chemical sense. Inside the little bumps on the top and sides of your tongue are 200 or more taste buds. Each contains a pore that catches food chemicals. These molecules are sensed by 50 taste receptor cells that project antenna-like hairs into the pore. Some of these receptors respond mostly to sweet-tasting molecules, others to salty, sour, or bitter-tasting ones.

(3) Although taste buds are essential for taste, there is more to taste than meets the tongue. Hold your nose, close your eyes, and have someone feed you various foods. A piece of apple may then be indistinguishable from a piece of raw potato; a piece of steak may taste like cardboard. To savor a taste, we normally breathe the aroma through our nose-which is why eating is not much fun when you have a bad cold, and why people who lose their sense of smell may think they have also lost their sense of taste. Smell not only adds to our perception of taste, it also changes it. A drink's strawberry odor enhances our perception of its sweetness. This is sensory interaction at work-the principle that one sense may influence another. Smell plus taste equals flavor.

Passage B: Smell

(1) Breaths come in pairs-inhale, exhale-except at two moments: birth and death. Each day, as you inhale and exhale nearly 20,000 breaths of life-sustaining air, you bathe your nostrils in a stream of molecules full of scents.

(2) Like taste, smell is a chemical sense. We smell something when air-carried molecules of a substance reach a tiny cluster of 5 million receptor cells at the top of each nasal cavity. These olfactory receptor cells respond selectively to the aroma of cake baking, to a wisp of smoke, to a friend's fragrance, and they instantly alert the brain through their axon fibers.

(3) Even nursing infants and mothers have a literal chemistry to their relationship, as they quickly learn to recognize each other's scents. Aided by smell, a mother fur seal returning to a beach crowded with pups will find her own. Our own sense of smell is less impressive than the acuteness of our seeing and hearing. Looking out across a garden we see its forms and colors in exquisite detail and hear its singing birds, yet smell little of it without jamming our nose into the flowers.

(4) Precisely how olfactory receptors work is a mystery. Unlike light, which can be separated into its spectral colors, an odor cannot be separated into more elemental odors. Thus, the olfaction system has no parallel to the retina, which detects myriad colors with sensory cells dedicated to red, green, or blue. Olfactory receptors recognize odors individually.

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الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

86 تفاصيل
What do the passages say about the receptor cells involved in taste and smell?

A Taste and smell require the use of at least 50 receptors.
B Taste and smell involve the use of 5 million receptor cells.
C Both the nose and the tongue have the same number of receptor cells.
D The receptor cells involved in smell are a lot more than those involved in taste.

87 تفاصيل
According to the passages, why are taste and smell important for us to enjoy food?

A There is more to taste than meets the tongue.
B Both senses interact to create flavors.
C Both senses have sense receptors.
D Both are chemical senses.

88 استنتاج
What can we understand from Passages A and B about both taste and smell?

A They are more impressive than hearing.
B They both involve the use of receptor cells.
C They both involve olfactory receptor systems.
D They both enhance the work of our other senses.

26 الإجابات مع الشرح

86 D The receptor cells involved in smell are a lot more than those involved in taste.

السبب في الأنف 5 ملايين خلية استقبال، وفي كل برعم تذوق 50 خلية فقط.

> *These molecules are sensed by 50 taste receptor cells ... a tiny cluster of 5 million receptor cells at the top of each nasal cavity*

87 B Both senses interact to create flavors.

السبب الشم والتذوق معًا يصنعان النكهة، لذلك يفقد الأكل طعمه مع الزكام.

> *Smell plus taste equals flavor.*

ليه الباقي غلط C و D صحيحان عن الحاستين لكنهما لا يشرحان «سبب الاستمتاع بالأكل».

ستيب STEP: تجميعات متكررة

88 **B They both involve the use of receptor cells.**

السبب القطعتان تذكران خلايا الاستقبال: خلايا التذوق وخلايا الشم.

> *These molecules are sensed by 50 taste receptor cells ... These olfactory receptor cells respond selectively*

ليه الباقي غلط A عكس القطعة B: الشم أضعف من البصر والسمع، و olfactory خاصة بالشم فقط.

ستيب STEP: تجميعات متكررة

27

الطيور العملاقة القديمة
Ancient Birds

ذكرها المختبرون 43 مرة

4 سؤال

(1) Giant birds of the past have names that speak for themselves. The Elephant bird, a native of Madagascar and the largest known giant bird, stood at over nine feet tall and weighed in at 1,000 pounds or more, until it went extinct about 1,000 years ago. Australia's Mthirung, nicknamed "Thunderbird," which disappeared nearly 50,000 years ago, is thought to have been nearly seven feet tall and weighed between 500 and 1,000 pounds. But until now, no one had ever found evidence of these towering avians in Europe.

(2) Today, researchers describe the first fossil of a giant bird found in Crimea. Dated at around 1.8 million years old, the specimen makes experts question previous assumptions that giant birds were not part of the region's fauna during that period.

(3) A team of paleontologists dug up the fossil, an unusually large femur bone, in Taurida Cave, located on the Crimean Peninsula in the northern Black Sea. The cave was only discovered last June when the construction of a new highway revealed its entrance.

(4) Based on the femur's dimensions, the team calculated that the bird would have weighed around 992 pounds as much as an adult polar bear-making it the third largest bird ever recorded. Daniel Field, a scientist at the University of Cambridge, says, "It raises exciting questions about the factors that gave rise to these giant birds, and the factors that drove them to extinction".

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

89 What two phrases does the writer use to say that the birds were very large?

تفاصيل

A giant bird and towering avian
B elephant bird and native of Madagascar
C large femur bone and the region's fauna
D seven feet tall and nicknamed "Thunder bird"

90 What can be understood about Daniel Field from paragraph (4)?

تفاصيل

A He studies fossil.
B He found the fossil.
C He weighed the fossil.
D He studies at Cambridge University.

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91 استنتاج

Which of the following does the writer say is a fact?

A The Thunderbird lived in Australia and Europe.
B The femur is the largest bone in the body.
C The Elephant bird was over nine feet tall.
D Researchers found a polar bear fossil.

92 استنتاج

How are researchers' views since the new discovery different from the views held before?

A They now know what the birds ate.
B They now know what the birds looked like.
C They now know when the birds became extinct.
D They now know a new place in which the birds lived.

27 الإجابات مع الشرح

89 A giant bird and towering avian

السبب الكاتب يصف ضخامة الطيور بعبارتي giant birds و towering avians.

Giant birds of the past ... no one had ever found evidence of these towering avians in Europe

90 A He studies fossil.

السبب Daniel Field عالم في جامعة كامبريدج يعلّق على الأحفورة، أي أنه يدرس الأحافير.

Daniel Field, a scientist at the University of Cambridge, says

ليه الباقي غلط الذي وجد الأحفورة فريق من علماء الحفريات، و D غلط لأنه عالم يعمل في الجامعة وليس طالبًا.

91 C The Elephant bird was over nine feet tall.

السبب طول طائر الفيل معلومة يذكرها الكاتب كحقيقة.

stood at over nine feet tall

92 D They now know a new place in which the birds lived.

السبب الاكتشاف في القرم (أوروبا) جعل الخبراء يراجعون رأيهم: الطيور العملاقة عاشت هناك أيضًا.

the specimen makes experts question previous assumptions that giant birds were not part of the region's fauna during that period

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28

طاقة الرياح
Wind Power (Passages A and B)

ذكرها المختبرون 43 مرة

2 سؤال

Passage A

(1) Wind power is an alternative energy source. This means that the power of the wind can be used in place of other energy sources such as coal, oil, and nuclear reactions. Wind can be used to produce electricity that heats homes and lights streets and buildings.

(2) Wind power is harnessed by a machine called a wind turbine. Wind turbines are tall towers topped with blades. The blades are connected to a vertical shaft, or rod. When wind causes the blades to spin, they turn the shaft. The turning shaft provides power to a machine called a generator, which produces electricity. A group of wind turbines, called a wind farm, can be used to provide electricity to a community. Small turbines provide power for individual homes.

(3) Wind turbines used to be called windmills. The earliest known windmills were built more than 1,000 years ago. People used them to grind grain. The Dutch used windmills to pump water out of low-lying areas near the sea.

Passage B

(1) Wind power is cost-effective. Land-based utility-scale wind is one of the lowest priced energy sources available today, costing 1-2 cents per kilowatt-hour after the production tax credit. Because the electricity from wind farms is sold at a fixed price over a long period of time (e.g. 20+ years) and its fuel is free, wind energy mitigates the price uncertainty that fuel costs add to traditional sources of energy.

(2) Wind creates jobs. The US wind sector employs more than 100,000 workers, and wind turbine technician is one of the fastest growing American jobs. According to the Wind Vision Report, wind has the potential to support more than 600,000 jobs in manufacturing, installation, maintenance, and supporting services by 2050.

(3) It's a clean fuel source. Wind energy doesn't pollute the air like power plants that rely on combustion of fossil fuels, such as coal or natural gas, which emit particulate matter, nitrogen oxides, and sulfur dioxide—causing human health problems and economic damages. Wind turbines don't produce atmospheric emissions that cause acid rain, smog, or greenhouse gases.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

93 What do Passages A and B say about wind?

تفاصيل

A Wind energy is free in some places.

B In the past, wind was used to crush grain.

C Wind energy is environmentally friendly.

D Wind can be used to produce electricity.

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استنتاج

94 **What can we understand from Passages A and B about wind farms?**

A A wind farm is easy to build and can provide power to a large number of people.

B Wind farms produce power more cheaply than individual wind turbines.

C Wind farms can provide both power and employment for a community.

D A wind farm consists of several wind turbines typically on a sea shore.

28 الإجابات مع الشرح

93 **D Wind can be used to produce electricity.**

السبب القطعتان تذكران أن الرياح تُستخدم لإنتاج الكهرباء.

> *Wind can be used to produce electricity that heats homes and lights streets and buildings.*

ليه الباقي غلط B و C كل واحد منهما في قطعة واحدة فقط، والسؤال عن القطعتين.

94 **C Wind farms can provide both power and employment for a community.**

السبب القطعة A: مزرعة الرياح تمد المجتمع بالكهرباء، والقطعة B: الرياح توفر وظائف.

> *A group of wind turbines, called a wind farm, can be used to provide electricity to a community. ... Wind creates jobs.*

ستيب STEP: تجميعات متكررة

29 بيتزا مارغريتا
Pizza Margherita

ذكرها المختبرون 43 مرة
2 سؤال

Some historians consider one man, an Italian pizzeria chef named Rafaelle Esposito, to be the creator of modern pizza. In 1889, he made a special pizza for Queen Margherita with tomatoes, mozzarella cheese, and fresh basil (to represent the red, white, and green colors of the Italian flag) that the queen highly enjoyed. That is how the original and now world-renowned pizza Margherita was born and paved the way for pizza to change from being a peasant's food to a popular dish for everyone throughout Italy and the rest of the world.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

تفاصيل

95 **Why do some historians consider Rafaelle Esposito as the founder of modern pizza?**

A The pizza he made was eaten by a Queen.
B The pizza he made used tomatoes from America.
C The pizza he made used the colors of the Italian flag.
D The pizza he made became a famous dish all over the world.

تفاصيل

96 **Which of the following is NOT true about pizza Margherita?**

A It is still considered a peasant's food today.
B It was first made by an Italian pizzeria chef.
C It is now a well-known dish around the world.
D It was designed especially for the queen of Italy.

29 الإجابات مع الشرح

95 **D The pizza he made became a famous dish all over the world.**

السبب: بيتزاه صارت مشهورة عالميًا وفتحت الطريق لانتشار البيتزا في كل العالم.

That is how the original and now world-renowned pizza Margherita was born and paved the way for pizza to change from being a peasant's food to a popular dish for everyone

ليه الباقي غلط: A و C صحيحان عن قصة البيتزا لكنهما ليسا سبب اعتباره مؤسس البيتزا الحديثة.

96 **A It is still considered a peasant's food today.**

السبب: المطلوب الخيار «غير الصحيح»: القطعة تقول إنها تحولت من أكل الفقراء إلى طبق للجميع.

to change from being a peasant's food to a popular dish for everyone

ستيب STEP تجميعات متكررة

30 السياحة في جامايكا
Tourism in Jamaica

ذكرها المختبرون 41 مرة
4 سؤال

(1) Tourism has become an increasingly important part of Jamaica's economy in recent decades. Tourism has improved the country's economy and increased its employment. It has also brought many opportunities to Jamaica's population. For example, in 2012 tourism provided almost $1.3 billion to Jamaica's economy.

(2) The income from tourism plays a major part in helping Jamaica's government to fund economic and social policies. The government has used the taxes from tourism to support special programs in education and health care. Tourism has also helped the country to improve its infrastructure or the basic system that a county needs to work property. One example of this is that Jamaica began to develop its roads and airports, which tourists use during their visit to the country.

(3) Along with developing the country's infrastructure, Jamaica has also started to protect some of its valuable natural resources. One example is that the country began to identify and protect its national parks and beaches. Now visitors must pay entry fees to visit these parks, which help to provide money to keep the parks protected. The government also works to keep the parks and beaches free from pollution and trash. These government policies have had a positive effect on tourism because tourists are able to have a better experience when they visit these places.

(4) Jamaica continues to rely heavily on tourism for its economy. However, one of the disadvantages of tourism in the country is the rainy season, which extends from May to November. During this time, 25% of the hotel workers lose their jobs because there are very few tourists. This has had a negative effect on the lives of the people who work in the tourism industry in the country.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

97 What did tourism lead to in Jamaica? تفاصيل

- A entry fees to parks.
- B increases in the prices of houses.
- C $1.3 billion added to the economy.
- D unemployment of 25% of hotel employees.

98 What happened to workers that depended on tourism? تفاصيل

- A They receive better wages and more leisure time.
- B They receive better health benefits from their employers.
- C They pay taxes that are used to improve roads and airports.
- D They are unemployed for seven months when there are few tourists.

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معنى كلمة

99 The word “major” in paragraph (2) is closest in meaning to _____

A important B basic C tiny D secondary

معنى كلمة

100 The word “identify” in paragraph (3) is closest in meaning to _____

A name B destroy C relax D sell

30 الإجابات مع الشرح

97 **C $1.3 billion added to the economy.**

السبب في 2012 أضافت السياحة قرابة 1.3 مليار دولار للاقتصاد.

in 2012 tourism provided almost $1.3 billion to Jamaica's economy

ليه الباقي غلط رسوم الحدائق سياسة حكومية، وفقدان الوظائف بسبب موسم الأمطار وليس بسبب السياحة نفسها.

98 **D They are unemployed for seven months when there are few tourists.**

السبب موسم الأمطار من مايو إلى نوفمبر (7 أشهر) يفقد فيه ربع عمال الفنادق وظائفهم.

the rainy season, which extends from May to November. During this time, 25% of the hotel workers lose their jobs

99 **A important**

السبب plays a major part = له دور مهم.

The income from tourism plays a major part in helping Jamaica's government

100 **A name**

السبب identify هنا = يحدد ويسمّي الحدائق والشواطئ ليحميها.

the country began to identify and protect its national parks and beaches

ستيب STEP: تجميعات متكررة

31 رسالة إلى توم
Message to Tom

ذكرها المختبرون 41 مرة
1 سؤال

"Steve: Can I come and see you next weekend?
Tom: on Friday I am busy, but Saturday will be OK."

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

تفاصيل

101 What did Steve ask in his message to Tom?

- A Are you usually busy at weekends?
- B What do you usually do on Fridays?
- C What are you going to do on Saturday?
- D Can I come and see you next weekend?

31 الإجابات مع الشرح

101 D Can I come and see you next weekend?

السبب: سؤال Steve مكتوب حرفيًا في الرسالة.

Can I come and see you next weekend?

ستيب STEP: تجميعات متكررة

32 استبيان وسائل التواصل (جدول)
Social Media Survey (table)

ذكرها المختبرون 39 مرة
2 سؤال

What type of social media do you use?

	Men	Women	Total
Twitter	34	42	76
Instagram	36	41	77
Linked in	11	3	14
Our square	9	6	15
Facebook	24	36	60

How many hours a day do you spend on social media?

	Men	Women	Total
0-1	2	3	5
1-3	17	24	41
4-6	19	28	47
7+	5	2	7

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

102 Which form of social media is least popular among women? — تفاصيل

A Twitter B Linked in C Instagram D Facebook

103 Which form of the social media is the most popular? — تفاصيل

A Twitter B Linked in C Instagram D Facebook

32 الإجابات مع الشرح

102 B Linked in

السبب: أقل رقم في عمود النساء هو 3 = Linked in.

Linked in: Men 11, Women 3, Total 14

103 C Instagram

السبب الأكثر شعبية = أعلى رقم في عمود Total، وهو Instagram = 77، ثم Twitter = 76.

> *Instagram: Men 36, Women 41, Total 77*

ستيب STEP: تجميعات متكررة

33

التشريح ووظائف الأعضاء
Human Anatomy and Physiology (Passages A and B)

ذكرها المختبرون 38 مرة

3 سؤال

Passage A: Human Anatomy

(1) Human anatomy is the scientific study of the body's structures. It is significant so we know how the structures or organs are supposed to work correctly. Some of those structures are very small and can only be observed and analyzed with the assistance of a microscope. Other larger structures can readily be seen, manipulated, measured, and weighed. The word "anatomy" comes from a Greek root that means "to cut apart." Human anatomy was first studied by observing the exterior of the body and observing the wounds of soldiers and other injuries. Later, physicians were allowed to dissect, or cut open, bodies of the dead to increase their knowledge.

(2) When a body is dissected, its structures are cut apart in order to observe their physical attributes and their relationships to one another. Dissection is still used in medical schools, anatomy courses, and in pathology labs. In order to observe structures in living people, however, a number of imaging techniques have been developed. These techniques allow clinicians to see structures inside the living body such as a cancer or a fractured bone.

(3) Like most scientific disciplines, anatomy has areas of specialization. Gross anatomy is the study of the larger structures of the body, those visible without the aid of magnification. Macro- means "large," thus, gross anatomy is also referred to as macroscopic anatomy. In contrast, micro- means "small," and microscopic anatomy is the study of structures that can be observed only with the use of a microscope or other magnification devices.

Passage B: Human Physiology

(1) Whereas anatomy is about structure, physiology is about function. Human physiology is the scientific study of the chemistry and physics of the structures of the body and the ways in which they work together to support the functions of life. It began in France over 200 years ago with a scientist named Claude Bernard. Much of the study of physiology centers on the body's tendency toward homeostasis. Homeostasis is the state of steady internal conditions maintained by living things. The study of physiology certainly includes observation, both with the naked eye and with microscopes, as well as manipulations and measurements. However, current advances in physiology usually depend on carefully designed laboratory experiments that reveal the functions of the many structures and chemical compounds that make up the human body and establish the foundation upon which we build our knowledge of how to treat disease.

(2) Like anatomists, physiologists typically specialize in a particular branch of physiology. For example, neurophysiology is the study of the brain, spinal cord, and nerves and how these work together to perform functions as complex and diverse as vision, movement, and thinking. Physiologists may work from the organ level (exploring, for example, what different parts of the brain do) to the molecular level (such as how an electrochemical signal travels along nerves).

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(3) Form is closely related to function in all living things. For example, the thin flap of your eyelid can snap down to clear away dust particles and almost instantaneously slide back up to allow you to see again.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

الفكرة الرئيسية

104 **Which of the following best summarizes anatomy and physiology from Passages A and B?**

A Anatomy involves cutting open, while physiology uses lab experiments.
B Anatomy is helpful for soldiers, while physiology is helpful for everyone.
C They both involve the study of neurophysiology.
D They both rely on cutting open bodies.

تفاصيل

105 **According to Passages A and B, it is important to know about anatomy and physiology _____**

A to be able to treat disease
B to be able to see cancer and find a cure
C to know how organs work and to treat disease
D to become a physiologist and a successful doctor

تفاصيل

106 **What do Passages A and B say about the study of anatomy and physiology?**

A Both mean to "cut apart"
B Both involve the use of microscopes
C Both are about the body's structure
D Both are taught at medical schools and high schools

33 الإجابات مع الشرح

104 **A Anatomy involves cutting open, while physiology uses lab experiments.**

السبب التشريح يعتمد على تقطيع الجسم وفحصه، وعلم وظائف الأعضاء يعتمد اليوم على تجارب المختبر.

Later, physicians were allowed to dissect, or cut open ... current advances in physiology usually depend on carefully designed laboratory experiments

105 **C to know how organs work and to treat disease**

السبب القطعة A: لنعرف كيف تعمل الأعضاء، والقطعة B: أساس معرفة علاج الأمراض.

It is significant so we know how the structures or organs are supposed to work correctly. ... establish the foundation upon which we build our knowledge of how to treat disease

ستيب STEP تجميعات متكررة

106 **B Both involve the use of microscopes**

السبب القطعتان تذكران استخدام المجهر.

> *can only be observed and analyzed with the assistance of a microscope ... both with the naked eye and with microscopes*

ليه الباقي غلط A و C خاصة بالتشريح فقط، والمدارس الثانوية غير مذكورة.

ستيب STEP: تجميعات متكررة

34 فريق روبيكون
Team Rubicon

ذكرها المختبرون 37 مرة
3 سؤال

Team Rubicon: Working for a not-for-profit organization

(1) Not-for-profit organizations contribute to a large part of economic activity in the United States. They are clearly very powerful in the business world, even though their focus on goals other than profit falls outside the traditional model of a for-profit business. But it is this move away from a focus on profit that allows them to follow their aims of social improvement and contribute to society as a whole. To be truly useful in a not-for-profit organization, a person must share the organization's vision.

(2) The vision for Team Rubicon was created by its founders, Jake Wood and William McNulty, who saw the devastation caused by the Haiti earthquake in 2010 and immediately took action. Both former soldiers, Wood and McNulty knew they could do something to help in this terrible situation. Within 24 hours, they had found the help of six other former soldiers, collected money, food and medicines from friends and family, and made their way to Haiti to help, and Team Rubicon was born.

(3) The organization gets its name from the Rubicon, a river in northern Italy that Julius Caesar and his army crossed on their historical march to Rome, with the river showing the point of no return. The name mirrors the founders' experiences during the Haitian disaster, where despite advice from the government and other aid organizations not to go ahead, their small team crossed into Haiti from the Dominican Republic carrying essential equipment and medicines to thousands of earthquake victims.

(4) Seven years later, Team Rubicon has two main aims: to pair the skills and experiences of former soldiers with emergency workers to be ready for any type of disaster and to provide a sense of community and achievement to former soldiers who have served their country but may be experiencing problems as a result of their war experiences.

(5) According to the organization's mission statement, Team Rubicon aims to provide former soldiers with three things that they sometimes lose after leaving the army: a purpose, gained through disaster relief; a sense of community, built by working with others; and a feeling of self-worth from recognizing the positive effect an individual can make when helping others.

(6) With its head office in the Los Angeles area, Team Rubicon has more than 60 employees who work in 10 regions around the country, along with more than 40,000 volunteers who are ready to move into action within 24 hours. Similar to company operations in for-profit organizations, staff positions at Team Rubicon include local administrators; field operations (including membership and training); marketing, communications, and social media; fundraising and partnership development; finance and accounting; and people operations.

(7) Team Rubicon's staff members bring professional and/or military experience to their daily jobs, but they all share the organization's vision. Many staff members started as volunteers for Team Rubicon while working in for-profit careers, while others took advantage of the organization's

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strong training program to become familiar with its aims.

(8) In 2016, Team Rubicon trained 8,000 former soldiers and emergency workers in disaster relief and responded to 16 disasters, which required more than 85,000 volunteer hours. In addition to donations from individuals, companies and other organizations, Team Rubicon relies on its partnerships with companies, such as Southwest Airlines, which provides hundreds of free plane tickets each year to fly volunteers to disaster sites.

(9) Team Rubicon engages its nationwide community at every level of the organization, from volunteer to board member, and every step of its operation: from training to planning to implementation to looking for and finding donations and volunteers to help with any type of disaster relief. Over the past several years, Team Rubicon has been recognized as one of the top non-profits to work for by The Non-Profit Times, based on employee surveys and business partners' opinions about the organization's work environment.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

تفاصيل

107 In 2016, how many disasters did Rubicon respond to?

A 10 B 24 C 16 D 60

تفاصيل

108 Where is Rubicon's head office?

A Dominican Republic B New York
C Haiti D Los Angeles

تفاصيل

109 From what country did Team Rubicon cross into Haiti?

A Dominican Republic B United States
C Cuba D Italy

34 الإجابات مع الشرح

107 C 16

السبب: في 2016 استجاب الفريق لـ 16 كارثة.

In 2016, Team Rubicon trained 8,000 former soldiers and emergency workers in disaster relief and responded to 16 disasters

108 D Los Angeles

السبب: المقر الرئيسي في منطقة لوس أنجلوس.

With its head office in the Los Angeles area

109 **A** Dominican Republic

السبب عبروا إلى هايتي من جمهورية الدومينيكان.

their small team crossed into Haiti from the Dominican Republic

ستيب STEP: تجميعات متكررة

35

خطر الطيران
The Risk of Flying

ذكرها المختبرون 36 مرة

3 سؤال

(1) There's a lot more to trying to evaluate whether a particular risk is big or small than you might think. Flying in airplanes is a case in point.

(2) You'd think that you could determine the numbers and the odds and that would be it. The annual risk of being killed in a plane crash for the average American is about 1 in 11 million. On that basis, the risk looks pretty small. Compare that, for example, to the annual risk of being killed in a motor vehicle crash for the average American, which is about 1 in 5,000.

(3) But if you think about those numbers, problems occur right away. Some people fly more and some fly less and some don't fly at all. So if you take the total number of people killed in commercial plane crashes and divide that into the total population, the result, the risk for the average American, may be a good general guide to whether the risk is big or small, but it's not specific to your personal risk.

(4) Then there is another numbers problem: You can calculate the risk of flying by dividing the number: (1) Of people who die into the total number of people, which gives you the risk for the average person; (2) Of victims into the number of total flights all passengers took, which gives the risk per flight; and some don't fly at all. (3) Of victims into the total number of miles all of them flew, which gives you the risk per mile. So, if you take the total number of people killed in commercial plane crashes and divide that into the total population, the result, the risk for the average American, may be a good general guide to whether the risk is big or small, but it's not specific to your personal risk.

(5) They all produce accurate numbers, but which one is most relevant to you depends on your personal flying patterns. Some fliers take lots of short flights and some take longer ones. Since the overwhelming majority of the few plane crashes that do occur take place in connection with takeoffs and landings, the risk is less a matter of how far you fly and more a matter of how often. If you are a frequent flier, then the risk per flight means more. For occasional long-distance fliers, the risk per mile means more. A frequent, long-distance flier would want to consider both.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

110 What are two words that the writer uses to mean "Find out"?

معنى كلمة

A think / divide.
B take / compare.
C produce / consider.
D determine / calculate

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111 How is the writer's view about plane crash risk different from the general belief? استنتاج

A It is about 1 in 11 million.
B It depends on national averages.
C It depends on individual flying habits.
D It depends on the condition of the plane.

112 According to the passage, which of the following is a fact about most plane crashes? استنتاج

A They happen during take off and landing.
B They happen during take off and mid-flight.
C They happen mid-flight and during landing
D They happen during medium-and long-distance flights.

35 الإجابات مع الشرح

110 D determine / calculate

السبب: determine في الفقرة 2 و calculate في الفقرة 4، والاثنتان بمعنى «يحسب ويعرف».

> *You'd think that you could determine the numbers ... You can calculate the risk of flying*

111 C It depends on individual flying habits.

السبب: الكاتب يرى أن الخطر يختلف حسب عادات طيرانك أنت، لا الرقم العام.

> *which one is most relevant to you depends on your personal flying patterns*

ليه الباقي غلط: A و B هما الطريقة العامة التي ينتقدها الكاتب.

112 A They happen during take off and landing.

السبب: أغلب الحوادث القليلة تقع عند الإقلاع والهبوط.

> *the overwhelming majority of the few plane crashes that do occur take place in connection with takeoffs and landings*

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36

تاريخ علم التشريح
Anatomy (History of the Science)

ذكرها المختبرون 36 مرة

3 سؤال

Anatomy

(1) Anatomy is a field in the biological sciences concerned with the identification and description of the body structures of living things. Gross anatomy involves the study of major body structures by dissection and observation and in its narrowest sense is concerned only with the human body. "Gross anatomy" customarily refers to the study of those body structures large enough to be examined without the help of magnifying devices, while microscopic anatomy is concerned with the study of structural units small enough to be seen only with a light microscope. Dissection is basic to all anatomical research. The earliest record of its use was made by the Greeks, and Theophrastus called dissection "anatomy," from ana temnein, meaning "to cut up." Comparative anatomy, the other major subdivision of the field, compares similar body structures in different species of animals in order to understand the adaptive changes they have undergone in the course of evolution.

Gross Anatomy

(2) This ancient discipline reached its culmination between 1500 and 1850, by which time its subject matter was firmly established. None of the world's oldest civilizations dissected a human body, which most people regarded with superstitious awe and associated with the spirit of the departed soul. Beliefs in life after death and a disquieting certainty concerning the possibility of bodily resurrection further inhibited systematic study. Nevertheless, knowledge of the body was acquired by treating wounds, aiding in childbirth, and setting broken limbs. The field remained speculative rather than descriptive, though, until the achievements of the Alexandrian medical school and its foremost figure, Herophilus (flourished 300 BCE), who dissected human cadavers and thus gave anatomy a considerable factual basis for the first time. Herophilus made many important discoveries and was followed by his younger contemporary Erasistratus, who is sometimes regarded as the founder of physiology. In the 2nd century CE, Greek physician Galen assembled and arranged all the discoveries of the Greek anatomists (scientists who study anatomy), including with them his own concepts of physiology and his discoveries in experimental medicine. The many books Galen wrote became the unquestioned authority for anatomy and medicine in Europe because they were the only ancient Greek anatomical texts that survived the Dark Ages in the form of Arabic (and then Latin) translations.

(3) Owing to church prohibitions against dissection, European medicine in the Middle Ages relied upon Galen's mixture of fact and fancy rather than on direct observation for its anatomical knowledge, though some dissections were authorized for teaching purposes. In the early 16th century, the artist Leonardo da Vinci undertook his own dissections, and his beautiful and accurate anatomical drawings cleared the way for Flemish physician Andreas Vesalius to "restore" the science of anatomy with his monumental De humani corporis fabrica libri septem (1543; "The Seven Books on the Structure of the Human Body"), which was the first comprehensive and illustrated textbook of anatomy. As a professor at the University of Padua, Vesalius encouraged

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younger scientists to accept traditional anatomy only after verifying it themselves, and this more critical and questioning attitude broke Galen's authority and placed anatomy on a firm foundation of observed fact and demonstration.

(4) From Vesalius's exact descriptions of the skeleton, muscles, blood vessels, nervous system, and digestive tract, his successors in Padua progressed to studies of the digestive glands and the urinary and reproductive systems. Hieronymus Fabricius, Gabriello Fallopius, and Bartolomeo Eustachio were among the most important Italian anatomists, and their detailed studies led to fundamental progress in the related field of physiology. William Harvey's discovery of the circulation of the blood, for instance, was based partly on Fabricius's detailed descriptions of the venous valves.

(5) The new application of magnifying glasses and compound microscopes to biological studies in the second half of the 17th century was the most important factor in the subsequent development of anatomical research. Primitive early microscopes enabled Marcello Malpighi to discover the system of tiny capillaries connecting the arterial and venous networks, Robert Hooke to first observe the small compartments in plants that he called "cells," and Antonie van Leeuwenhoek to observe muscle fibres and spermatozoa. Thenceforth attention gradually shifted from the identification and understanding of bodily structures visible to the naked eye to those of microscopic size.

(6) The use of the microscope in discovering minute, previously unknown features was pursued on a more systematic basis in the 19th century, but progress tended to be slow until technical improvements in the compound microscope itself, beginning in the 1830s with the gradual development of achromatic lenses, greatly increased that instrument's resolving power. These technical advances enabled Matthias Jakob Schleiden and Theodor Schwann to recognize in 1838-39 that the cell is the fundamental unit of organization in all living things. The need for thinner, more transparent tissue specimens for study under the light microscope stimulated the development of improved methods of dissection, notably machines called microtomes that can slice specimens into extremely thin sections. In order to better distinguish the detail in these sections, synthetic dyes were used to stain tissues with different colors. Thin sections and staining had become standard tools for microscopic anatomists by the late 19th century. The field of cytology, which is the study of cells, and that of histology, which is the study of tissue organization from the cellular level up, both arose in the 19th century with the data and techniques of microscopic anatomy as their basis.

(7) In the 20th century anatomists tended to scrutinize tinier and tinier units of structure as new technologies enabled them to discern details far beyond the limits of resolution of light microscopes. These advances were made possible by the electron microscope, which stimulated an enormous amount of research on subcellular structures beginning in the 1950s and became the prime tool of anatomical research. About the same time, the use of X-ray diffraction for studying the structures of many types of molecules present in living things gave rise to the new subspecialty of molecular anatomy.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

113 تفاصيل

Who were among the first notable people to illustrate their studies of anatomy?

A Leonardo da Vinci and Andreas Vesalius
B Herophilus and Leonardo da Vinci
C Alexandrian and Herophilus
D Galen and Erasistratus

114 تفاصيل

What things did primitive microscopes allow Antonie van Leeuwenhoek to study?

A small compartments and muscle fibers
B muscle fibers and spermatozoa
C arterial and venous networks
D spermatozoa and eyes

115 تفاصيل

Which new technologies enabled anatomists to see details beyond the limits of light microscopes?

A increased resolving power and X-ray diffraction
B electron microscope and X-ray diffraction
C synthetic dyes and achromatic lenses
D dissection tools and transparency

36 الإجابات مع الشرح

113 **A Leonardo da Vinci and Andreas Vesalius**

السبب رسومات دافنشي الدقيقة، ثم كتاب فيزاليوس المصوّر.

> *Leonardo da Vinci undertook his own dissections, and his beautiful and accurate anatomical drawings ... the first comprehensive and illustrated textbook of anatomy*

114 **B muscle fibers and spermatozoa**

السبب ليفينهوك رأى ألياف العضلات والحيوانات المنوية.

> *Antonie van Leeuwenhoek to observe muscle fibres and spermatozoa*

ليه الباقي غلط الحجيرات الصغيرة cells رآها Robert Hooke، والشعيرات بين الشرايين والأوردة رآها Malpighi.

115 **B electron microscope and X-ray diffraction**

السبب المجهر الإلكتروني وحيود الأشعة السينية في القرن العشرين.

> *These advances were made possible by the electron microscope ... the use of X-ray diffraction*

ستيب STEP: تجميعات متكررة

37

تقنية النانو وأقدام الوزغ
Nanotechnology (Gecko Feet)

ذكرها المختبرون 36 مرة

3 سؤال

Nanotechnology

(1) Anybody who doesn't know much about nanotechnology should begin with geckos. These are the lizards that are probably the world's best climbers. Watching them climb upside down on a horizontal pane of glass, you realize that Spiderman should really have been called Geckoman. These guys out-climb spiders any day.

(2) With perfect ease they can hang from a single toe, and they do so by pure adhesion, not by sticking a toe in a hole or by curling it round something they can grip. With all the toes on the glass scientists guess that if the rest of the body were strong enough it could take the weight of a 100kg person suspended below it. Although each toe is equipped with a tiny hook like claw at the end, these are of no use on the glass. What keeps them up there is the amazing structure of the skin of the toe.

(3) Seen under the microscope each toe has around two million tiny hairs on its underside. Under the higher magnification of an electron microscope the end of each of these hairs is seen to split into hundreds of even tinier nano-hairs, which scientists have called spatulae. These hairs are so small that they are able to establish contact with the molecular structure of the surface the gecko is walking on. With that near-perfect contact the hairs are stuck to the surface by electromagnetic forces called Van der Waals forces. The molecules on the feet and on the surface have areas of slight positive or negative charge that attract each other like mini magnets when they get really close.

(4) Scientists have been working for over 15 years now to try to unlock the secrets of the stickiness of gecko toes and find a way to artificially reproduce the same structure of nano-hairs. The hypothesis at the moment is that if any material can be shaped into nano hairs they will have the same properties as those on gecko toes, so scientists are looking for a different material with which to make the stickiest synthetic surface ever.

(5) This is one example of research in the field now known as nanotechnology. In this field, the technology being created can be measured in a few nanometers (one nanometer is a millionth of a millimeter). Interest in developing technology at this level was largely inspired by Richard Feynman at the beginning of the 1960s, but the "nano" catchphrase was coined in the 1990s by Dr. Eric Drexler, who spurred scientists on through a series of speeches and a book with the title "Engines of Creation: The Coming Era of Nanotechnology"

(6) The scientists working on the geckos see an enormously wide range of possible applications for the kind of adhesive nanotechnology that they will develop. The one that will make the biggest splash in the media will be the gloves and the boots that will allow rockclimbers to take their sport to hitherto undreamed of heights. But the technology could also be used in surgery to keep the

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edges of wounds together without the need for stitches. There will also be a huge potential in the manufacturing sector to stick millions of components together tighter than ever before without glue or screws.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

116 How many hairs does a gecko have at the bottom of each toe? تفاصيل

A fifteen
B hundreds
C around two thousand
D around two million

117 What are the forces that allow geckos to climb upside down on glass called? تفاصيل

A Van der Waals forces
B Sticking forces
C Electric forces
D Spatulae forces

118 Who started using the popular phrase “nano”? تفاصيل

A Geckoman
B Eric Drexler
C Albert Einstein
D Richard Feynman

37 الإجابات مع الشرح

116 D around two million

السبب: في كل إصبع نحو مليوني شعرة.

each toe has around two million tiny hairs on its underside

ليه الباقي غلط: «مئات» هو عدد الشعيرات النانوية في طرف كل شعرة.

117 A Van der Waals forces

السبب: الشعيرات تلتصق بالسطح بقوى كهرومغناطيسية اسمها Van der Waals.

the hairs are stuck to the surface by electromagnetic forces called Van der Waals forces

118 B Eric Drexler

السبب: عبارة «نانو» صاغها Eric Drexler في التسعينيات.

the "nano" catchphrase was coined in the 1990s by Dr. Eric Drexler

ليه الباقي غلط: Feynman ألهم الاهتمام بالمجال في الستينيات، لكنه لم يصغ الكلمة.

ستيب STEP تجميعات متكررة

38

اضطرابات النوم
Sleep Disorders

ذكرها المختبرون 34 مرة

5 سؤال

Sleep Disorders

(1) The idea that "everyone needs 8 hours of sleep" is untrue. Newborns spend nearly two-thirds of their day asleep, most adults no more than one-third. Age-related differences in average time spent sleeping are rivaled by differences in the normal amount of sleep among individuals at any age. Some people thrive with fewer than 6 hours of sleep per night; others regularly sleep 9 hours or more. Sleep patterns may be genetically influenced. When two researchers, Wilse Webb and Scott Campbell, checked the pattern and duration of sleep among fraternal and identical twins, only the identical twins were strikingly similar.

(2) Whatever their normal need for sleep, some 10 to 15 percent of adults complain of insomnia—persistent problems in falling or staying asleep. True insomnia is not the occasional inability to sleep that we experience when anxious or excited. For a stressed organism, alertness is natural and adaptive. Moreover, from middle age on, sleep is seldom uninterrupted. Occasional awakenings become the norm, not something to fret over or treat with medication.

(3) Sometimes people fret unnecessarily about their sleep. Scientists have learned to doubt people's fretful sleep reports. In laboratory studies, insomnia complainers do get less sleep than others, but they typically overestimate, by about double, how long it took them to fall asleep, and they underestimate by nearly half how long they actually slept. Even if we've been awake only an hour or two, we may think we've had little sleep because it's the waking part we remember. When researchers awaken people repeatedly during the night, some recall having slept soundly.

(4) The most common quick fixes for true insomnia—sleeping pills and alcohol—can aggravate the problem. Both reduce REM sleep and can leave a person with next-day blahs. With continued use, one needs bigger doses to get an effect, and when the drug is discontinued, the insomnia may worsen. Scientists are searching for natural chemicals that are abundant during sleep and might be synthesized as a sleep aid without side effects. In the meantime, sleep experts offer other natural alternatives:

• Relax before bedtime, using dimmer light.
• Avoid caffeine (this includes chocolate) after late afternoon and avoid rich foods before bedtime. A glass of milk may help. (Milk provides raw materials for the manufacture of serotonin, a neurotransmitter that facilitates sleep.)
• Sleep on a regular schedule (rise at the same time even after a restless night) and avoid naps. A regular sleep schedule boosts daytime alertness, too, as shown in a recent experiment in which University of Arizona students slept 7.5 hours a night on either a varying or consistent schedule.
• Exercise regularly but not in the late evening (late afternoon is best).
• Reassure yourself that the temporary loss of sleep causes no great harm, certainly nothing worth losing sleep over.
• If nothing else works, aim for less sleep; go to bed later or get up earlier.

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(5) More rare but also more severe than insomnia are the sleep disorders narcolepsy and sleep apnea. People with narcolepsy (from narco, "numbness," and lepsy, "seizure") suffer periodic, overwhelming sleepiness. This usually lasts less than 5 minutes but sometimes occurs at the most inopportune times, perhaps just after taking a terrific swing at a softball or when laughing loudly or shouting angrily. In severe cases, the person may collapse directly into a brief period of REM sleep, with its accompanying loss of muscular tension. Those who suffer from narcolepsy—1 in 2000 people, estimates the Stanford University Center for Narcolepsy—must live with extra caution.

(6) The National Heart, Lung, and Blood Institute reports that 1 in 25 people (mostly overweight men) suffer from sleep apnea—a treatable disorder that was unknown before modern sleep research. They intermittently stop breathing during sleep. (Apnea means "stopping respiration.") After an airless minute or so, decreased blood oxygen arouses the sleeper to awaken and snort in air for a few seconds. The process can repeat more than 400 times a night, depriving the person of slow-wave sleep. Apart from complaints of sleepiness and irritability during the day—and their friends' complaints about their loud "snoring"—apnea sufferers are often unaware of their disorder.

الأسئلة
حلها بنفسك أولًا، والإجابات تحتها

119 What percentage of adults have problems falling asleep? تفاصيل

A 2-3 percent
B 6-8 percent
C 10-15 percent
D 20-25 percent

120 How many people suffer from narcolepsy? تفاصيل

A 1 in 5 people
B 1 in 25 people
C 1 in 400 people
D 1 in 2000 people.

121 Who did a study on sleep apnea? تفاصيل

A the university of Arizona
B the New York Center for Sleep Apnea
C The National Heart, Lung, and Blood Institute
D The Stanford University Center for Narcolepsy

122 Who did Wilse Webb and Scott Campbell study? تفاصيل

A different types of men
B different types of twins
C different types of adults
D different types of parents

123 Who usually experiences sleep apnea? تفاصيل

A fat men
B thin women
C normal adults
D identical twins

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38 الإجابات مع الشرح

119 **C 10-15 percent**

السبب من 10 إلى 15 بالمئة من البالغين يشتكون من الأرق.

> *some 10 to 15 percent of adults complain of insomnia*

120 **D 1 in 2000 people.**

السبب واحد من كل 2000 شخص.

> *1 in 2000 people*

121 **C The National Heart, Lung, and Blood Institute**

السبب المعهد الوطني للقلب والرئة والدم هو مصدر معلومة انقطاع النفس.

> *The National Heart, Lung, and Blood Institute reports that 1 in 25 people*

122 **B different types of twins**

السبب درسا التوائم المتماثلة وغير المتماثلة.

> *checked the pattern and duration of sleep among fraternal and identical twins*

123 **A fat men**

السبب أغلب المصابين رجال زائدو الوزن.

> *(mostly overweight men) suffer from sleep apnea*

ستيب STEP: تجميعات متكررة

39 أزمة المياه في البرازيل
Brazil's Water Shortage

ذكرها المختبرون 32 مرة
3 سؤال

(1) Brazil continues to suffer from a historic drought (lack of water), and millions of people in its largest city might run out of water soon. São Paulo, home to around 20 million people, is experiencing its lowest rainfall since 1930, and new water-saving measures have been introduced to deal with such a very difficult situation. 90% of people in São Paulo say that they are being affected by the water shortage. People in other cities such as Rio Janeiro with 7 million people and the capital city of Brasília with 5 million people are not experiencing the same problems that are happening in São Paulo.

(2) Services, including schools and hospitals, are having to adapt to the country's new water problems. The Cândido Fontoura children's hospital has denied that it ran out of water earlier this month, but biologist Analice Dora expressed fear: "Every one is worried. Hospitals are the one place that can't lack water." If there is not enough water in hospitals, people could die because they cannot get medical treatments. Water is important to hospitals because it is important for the area to be clean.

(3) Although there has been a recent increase in rainfall in the region, it hasn't been enough to improve supply in Brazil, a country which normally has 12 percent of the world's freshwater – more water than any other country in the world.

(4) current reserves in São Paulo stand at just 10 percent — known as the "dead volume" — and the government has warned that it could get worse in the next few months. Reserves are the water that is stored or kept. This is usually used in cases of emergencies.

(5) Professor Decio Semensatto from the Federal University of São Paulo has compared the current water situation to a "semi-desert." And the water crisis will probably last for years. He says that water utility Sabesp is responsible because it has known for years that drought-like conditions would soon arrive, but did not do much to prevent the future problems. The current crisis has reached a serious point, and water rationing may be necessary. Rationing is where only small amounts of something are available. People will only be able to get small amounts of water and will have to use it for only things that are very important. São Paulo State governor Geraldo Alckmin last month admitted that it was already happening. Brazil is not used to such scarcity, but São Paulo is getting used to the way things have to run; There are incentives to use less water, fines for those who use too much and the possible installation of more efficient taps. Sabesp will see if such steps can make the difference before it decides whether to formally introduce wider water rationing. President Bolsonaro has not done anything to help resolve the problem. He does not think that it is a problem for the federal government to solve and has said that the local government needs to fix the problem.

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الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

124 How many people live in Rio de Janeiro? تفاصيل

A 5 million B 7 million C 12 million D 20 million

125 Who is the governor of São Paulo? تفاصيل

A Decio Semensatto B Analice Dora
C Geraldo Alckmin D Roberto Sitoe

126 How many people in São Paulo say that they are being affected by the water shortage? تفاصيل

A 10% B 12% C 90% D 95%

39 الإجابات مع الشرح

124 B 7 million

السبب ريو دي جانيرو 7 ملايين.

Rio Janeiro with 7 million people

ليه الباقي غلط 20 مليون ساو باولو، و 5 ملايين برازيليا.

125 C Geraldo Alckmin

السبب حاكم ولاية ساو باولو هو Geraldo Alckmin.

São Paulo State governor Geraldo Alckmin

126 C 90%

السبب 90% من سكان ساو باولو يقولون إنهم متأثرون.

90% of people in São Paulo say that they are being affected by the water shortage.

ليه الباقي غلط 10% هي نسبة المخزون، و 12% نصيب البرازيل من المياه العذبة.

ستيب STEP: تجميعات متكررة

40 أنواع الضرائب
Taxes

ذكرها المختبرون 31 مرة
2 سؤال

(1) Consumption taxes: As the name implies, consumption taxes tax consumption. There are different variants of consumption taxes. Three important consumption taxes are the sales tax, the excise tax, and the value added tax (VAT).

(2) Sales Tax: A sales tax is a tax on sales and are usually a flat percentage of the amount of the purchase. Sellers usually collect sales taxes at the point of sale from the ultimate consumer. Many states and localities rely on sales tax for a substantial portion of their revenue needs. Sales taxes are relatively easy to collect. By their very nature, sales taxes are not collected on amounts that citizens or residents save. Hence, their effect is more burdensome to those persons who must spend more (even all) of their income to purchase items subject to a sales tax. While such taxes are nominally in equal percentage of all purchases, their effect is regressive for those who accumulate wealth and who spend all of their income on items subject to them.

(3) In states that have sales taxes applicable to all purchases, every citizen or resident who buys anything pays more sales tax. In this sense, citizen beneficiaries may more equitably share the burden of paying for state or local government than is the case of the financing schemes already noted. The recent financial crisis has made clear that a state's revenue are vulnerable to economic downturns during which citizens or residents must reduce their purchases. Such downturns are the very occasions when states need more funds to finance services for which their citizen stand in greater need.

(4) Sales taxes are particularly attractive to states that perceive an ability to pass them to a non-citizens or non-residents. There is nothing quite so politically attractive as making someone who cannot vote in state elections increase the state's funds. Tourist-destination states that persons from out-of-state visit find sales taxes attractive.

(5) Excise taxes: An excise tax is a sales tax that applies only to certain classes of good, e.g., luxury items. Excise taxes on luxury items may be politically popular, but those excise taxes do not raise much revenue because they are avoidable. The demand for luxury items is usually limited. Excise taxes on high-demand (arguably) non-necessities, e.g., cellular telephones, raise much more revenue. Tourist-destination states find excise taxes on services that old-of-state visitors are more likely to purchase than residents to be attractive e.g., renting cars, staying in hotels, visiting tourist sites.

(6) Some states impose excise taxes on "sin" purchases e.g., cigarettes. The public health costs associated with activities such as smoking or drinking may be high, so states tax heavily the purchase of products that cause it to have to provide costly health services. Arguably, such taxes may discourage persons from making the purchases in the first place.

(7) Value Added Tax: This tax is imposed upon every sale, not only the sale to the ultimate consumer, i.e., it is imposed at every stage of production of a product. The seller pays the VAT to the government minus whatever tax the seller was assessed upon acquiring the good. Thus, the

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tax base is only a purchaser's actual additions to the value of a product. Since the final consumer does not resell the product, he/she/it pays the final tax bill. Many European countries favor a VAT, often in combination with an income tax.

(8) There are of course other forms of taxation used by nations to raise income. These include wealth taxes and wage taxes.

(9) Wealth Taxes: We could tax wealth. There are at least two common forms of wealth taxes, estate taxes and property taxes. The estate tax is imposed on the estates of the deceased and the amount of the tax depends on the size of the estate. Property taxes are imposed on taxpayers because they own property. Municipalities often rely on property taxes to raise the revenues they need. Notices that in the case of property taxes, the taxing authority can tax the same "wealth" again and again, e.g., every year.

(10) Wage taxes: we could tax wages by a flat percentage irrespective of how much those wages are. This is sometimes called a "payroll" tax. Some states rely on payroll tax. It is cheaper to administer than an income tax because there are few deductions or exclusions from the tax base. Social Security taxes and Medicare taxes are wage taxes. The tax rate of the Social Security Tax (6.2% on both employer and employee) is limited to the first $120.000 of wage income. The maximum on the tax rate of the Social Security Tax of course creates an effect where those with incomes higher than the maximum pay a rate that is lower than the rate of those whose income is below the maximum rate of the Medicare tax (1.45% on both employer and employee) is not subject to a limit. Therefore, for high-income earners, there is an additional 0.9% tax on wages or self-employment income.

الأسئلة

حلها بنفسك أولًا، والإجابات تحتها

127 تفاصيل

According to the passage, what are kinds of wage taxes?

A income tax and social security tax
B social security tax and medicare taxes
C purchase and medicare taxes
D payroll and wealth taxes.

128 تفاصيل

According to the passage, which of the following are kinds of excise tax?

A cigarettes and property
B property and wages
C cigarettes and wages
D cellphones and cigarettes

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40 الإجابات مع الشرح

127 **B** social security tax and medicare taxes

السبب القطعة تقول صراحة إن الضمان الاجتماعي والرعاية الطبية ضرائب أجور.

> *Social Security taxes and Medicare taxes are wage taxes.*

ليه الباقي غلط ضريبة الدخل والضريبة على الثروة نوعان مختلفان.

128 **D** cellphones and cigarettes

السبب الجوالات مثال لضريبة إنتاج على سلع مطلوبة، والسجائر مثال لضريبة «الخطيئة».

> *Excise taxes on high-demand (arguably) non-necessities, e.g., cellular telephones ... Some states impose excise taxes on "sin" purchases e.g., cigarettes.*

ليه الباقي غلط العقار والأجور لها ضرائب أخرى (الثروة والأجور).

ستيب STEP: تجميعات متكررة

اتدرب على كل القطع في التطبيق

ستيب STEP: تجميعات متكررة

- أكثر من 2000 سؤال من تجميعات المختبرين: قراءة، قواعد، استماع، وتحليل كتابي.
- شرح عربي لكل سؤال، ودروس فيديو للتأسيس من الصفر.
- اختبارات تجريبية بنفس شكل الاختبار، ومتابعة للأسئلة اللي أخطأت فيها.
- تحديث مستمر بآخر ما يذكره المختبرون.

للآيفون: من App Store

https://apps.apple.com/sa/app/id6821437506

للأندرويد: من Google Play

https://play.google.com/store/apps/details?id=com.steprecurring.app

امسح الرمز بكاميرا الجوال، أو افتح الرابط اللي تحته.

قناة التليجرام: اختبار يومي مجاني الساعة ٨ مساءً

امسح الرمز وانضم للقناة، أو افتح الرابط:

t.me/step_tajmeeat

شارك الملف مع زملائك، وبالتوفيق في اختبار ستيب.